ReTell: Supporting Children with ADHD in Literacy Learning through a Mobile, LLM-Powered Interactive Task-Based Learning System

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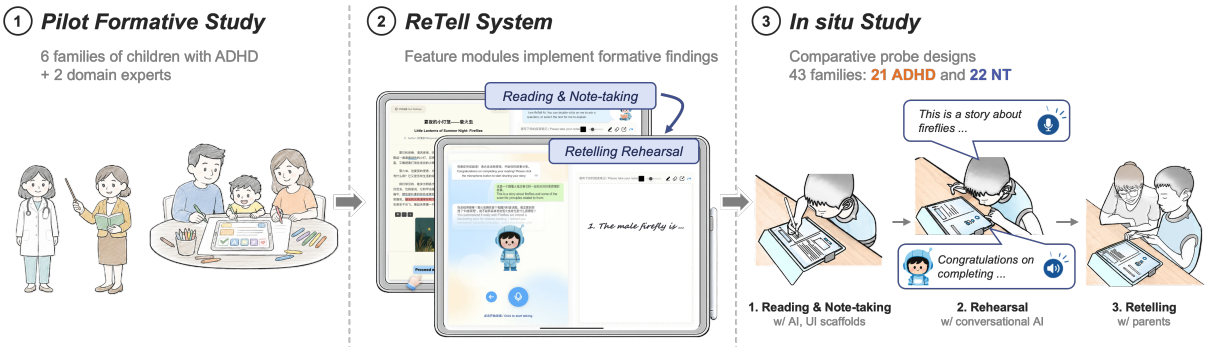

Fig. 1. Overview of our research process. (1) A pilot formative study with 6 families of children with ADHD and 2 domain experts informed (2) ReTell, a tablet app that guides children through reading, note taking, and retelling rehearsal. (3) An in situ study with comparative probe designs involving 43 families (21 ADHD, 22 neurotypical).

For children with Attention-Deficit/Hyperactivity Disorder (ADHD), their Executive Function Deficits (EFDs) can undermine sustained attention and make literacy tasks particularly demanding. Task-Based Learning (TBL) structures learning as a

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sequence of manageable subtasks leading toward a meaningful outcome, making it potentially well suited to the attentional profiles of children with ADHD. However, TBL practice often relies on expert orchestration, and how to operationalize its principles as an interactive system for everyday use remains unclear. We present ReTell, an LLM-organized TBL system that supports reading, note-taking, and retelling. Informed by a pilot formative study with six families of children with ADHD and two domain experts, ReTell organizes LLM support within a structured, context-aware workflow that provides specific scaffolding across the three phases. We evaluated ReTell through an in situ comparative probe study involving 43 families (21 ADHD and 22 neurotypical). No statistically significant differences between children with ADHD and their neurotypical peers were detected in content coverage or comprehension during the retelling sessions, while interaction logs and interviews revealed distinct support needs and possible compensatory strategies, such as more frequent revisits to the article. Grounded in these findings, we derive design implications for adaptive pacing, external memory, mixed-initiative assistance, and accessible multimodal input in ubiquitous neuro-inclusive learning systems.

CCS Concepts: • **Human-centered computing → Empirical studies in HCI**; • **Social and professional topics → Children**; • **Applied computing → E-learning**.

Additional Key Words and Phrases: ADHD, task-based learning, mobile app, large language model

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1 Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is increasingly prevalent among school-aged populations, with incidence peaking at ages 5-9 and prevalence highest at ages 10-14 [25, 119]. Beyond behavioral hyperactivity, children with ADHD face profound Executive Function Deficits (EFDs) that impair sustained attention, working memory, and the self-organization of complex information [9, 106, 116]. These deficits render foundational literacy tasks, such as reading comprehension and structured narrative expression, exceptionally demanding and often lead to diminished academic well-being [15, 26, 49, 63, 71].

To address these challenges, one effective pedagogical framework is Task-Based Learning (TBL). Its core mechanics, i.e., chunking complex goals into manageable units and providing iterative feedback [118], also echo well with established clinical interventions for ADHD, such as Organizational Skills Training (OST), Daily Report Cards (DRCs), and relevant behavioral management training [16, 44, 83]. While these TBL-aligned interventions are well-established across classroom and clinical settings [10, 27], their successful execution remains expert-dependent, requiring specialized pedagogical knowledge and complex organizational orchestration [32, 109]. In home-based settings, where children spend the majority of their learning activities, this intellectual and organizational labor falls on parents, creating a support gap where the children in need are the least likely to receive such services [33, 45].

We are witnessing the rapid development of Large Language Models (LLMs), AI agents that offer a transformative opportunity to provide ubiquitous, specialized support for ADHD learners, owing to their vast knowledge bases, 24/7 availability, and capacity for patient, natural-language interaction [14, 48, 61]. However, normal LLM-based interfaces also pose significant risks of misuse and cognitive misalignment for such populations. In the context of literacy learning, open-ended chatbots can be misused to provide immediate solutions that bypass the child's internal synthesis and undermine the development of agency (namely, cognitive offloading, choice overload) [12, 43, 50, 58]. Furthermore, the infinite flexibility of these interfaces implicitly assumes a baseline of executive control that ADHD learners unfortunately struggle to maintain; without explicit structural constraints, the interaction itself could become a burden rather than a support [76, 111]. To address these challenges, the community is advocating for systematic, neuro-inclusive Human-AI Interaction (HAI) design that replaces

open-ended dialogue with structured workflows to ensure cognitive support without compromising the learning process [120]. From the systems perspective, the open question is therefore not whether LLMs can converse helpfully with children, but how to architect the orchestration layer around them: how phase transitions, context injection, initiative, and logging should be structured so that an evidence-based intervention retains its fidelity when it leaves expert hands and enters unsupervised home use. This is an interactive systems problem: it concerns interaction architecture, in situ deployment, and behavioral instrumentation rather than curriculum content. Therefore, this paper aims to answer two primary research questions: **RQ1**: How can TBL be architected into an LLM-organized app accessible to children with ADHD? **RQ2**: How do children with ADHD interact with this system, particularly in comparison with their neurotypical (NT) peers, and what are the design implications in promoting ADHD-specific learning?

To address these, we present ReTell (**Re**ading with note-taking to re**Tell**), an LLM-organized TBL app for structured literacy learning designed with ADHD families and domain experts. The app is anchored in *retelling* [89]: after reading a passage, the child first rehearses re-expressing its content in their own words with an AI agent through scaffolded, step-wise guidance, and then delivers the final retelling to their parent as an authentic audience. This design was recommended by domain experts in our pilot formative study as a developmentally appropriate anchor task for children with ADHD. It also aligns with both forms of TBL [30]: note-taking-supported reading serves as the input phase [64], while retelling operationalizes the output-based principle, akin to the Feynman Technique [88], where articulating knowledge in one's own words reveals and consolidates understanding (Section 2). In an in situ study using comparative probe designs, we detected no statistically significant differences in retelling performance between children with ADHD and their NT peers during the sessions, even though parents reported weaker literacy habits for children with ADHD at home. Because the study included no controlled comparison condition, we treat this as a descriptive observation of how children performed within the ReTell workflow rather than as evidence that ReTell caused these outcomes; our probe design also evaluates ReTell as an integrated whole and does not isolate the contribution of any single component. This paper makes the following contributions:

(1) **A traceable design process.** We conducted a three-stage, multi-stakeholder formative study whose findings are traceably mapped onto design goals and system features, offering a derivation template that may be reused for building interactive systems with and for neurodivergent children.
(2) **A system architecture for LLM orchestration.** We designed and implemented ReTell, a tablet-based system that rearchitects open-ended LLM interaction into a phase-organized workflow to support literacy learning of children, especially those with ADHD.
(3) **In situ empirical evaluation.** We conducted a comparative probe study with 43 families (21 ADHD, 22 NT), detecting no statistically significant group differences in retelling performance during the sessions and identifying interaction patterns specific to children with ADHD, from which we distill four design implications for interactive and ubiquitous learning systems.

2 Related Work

2.1 Learning Challenges for Children with ADHD

Children with ADHD often encounter significant difficulties when engaging in learning tasks that require sustained and structured cognitive processing [9, 116]. Beyond behavioral hyperactivity, these children exhibit impaired temporal processing and a diminished capacity to maintain coherent mental representations in working memory over extended periods [68, 85]. Thus, they frequently struggle with high-level, long-cycle tasks in which knowledge must be progressively constructed, integrated, and externalized [29]. This makes foundational literacy tasks such as reading comprehension and structured narrative expression particularly demanding [49, 71, 106].

Generally, these executive function deficits compound over time, often resulting in diminished academic well-being and long-term educational disadvantages [15, 26, 63].

To mitigate these challenges, children with ADHD often require externalized scaffolds to compensate for internal executive function deficits [32]. In clinical and classroom settings, current practices employ a variety of structured programs: Organizational Skills Training (OST) [2] focuses on remediating deficit skills by establishing systematic routines and checklists for task management, while Daily Report Cards (DRC) [44, 113] function as a behavioral intervention tool that translate overarching expectations into monitored behavioral targets to provide immediate feedback throughout the day. However, because these interventions mostly occur in professional settings, their ecological validity is often compromised by the transition to home settings. The complexity of managing these interventions often exceeds parents' capacity, as they struggle to balance the roles of caregiver and clinician [33, 45]. This pedagogical overhead makes it difficult to replicate the structured scaffolding of expert-led environments in the dynamic and often chaotic context of the everyday family environment, leaving the children with ADHD who most need structured support the least likely to receive it. For our community, the gap is not a lack of intervention knowledge but the lack of a practical, ubiquitous approach to translating it into daily use with its structure intact.

2.2 Task-Based Learning in Children's Education

Task-Based Learning (TBL) is a pedagogical framework that structures learning around meaningful, goal-oriented tasks [118]. Also referred to as Task-Based Language Learning (TBLL), it engages learners across the literacy domains of listening, reading, writing, and speaking [81]. Beyond its origins in English as a Foreign Language (EFL) teaching, TBL has evolved into a versatile approach that can be tailored to meet children's diverse learning needs, in part because children's first-language (L1) acquisition shares similar cognitive routes with adults' second-language (L2) learning [39]. The success of TBL depends on designing tasks that motivate engagement and present clear purposefulness [55, 91].

In practice, TBL manifests in two complementary forms [30]. Output-based TBL centers on production-oriented tasks such as storytelling, role-plays, or collaborative projects that require learners to actively generate language and negotiate meaning [31, 118]. This process enables learners to recognize gaps in their knowledge and refine them through interaction and feedback [96, 105, 110], similar to the Feynman Technique [88]. Among these output practices, orally reconstructing a text in one's own words, namely *retelling*, is a canonical strategy in literacy education for improving comprehension and narrative structure [73]. Input-based TBL, by contrast, prioritizes comprehension through carefully designed listening or reading activities, where elaborated input that is contextually rich yet cognitively accessible has proven most effective [64].

Critically, the core mechanics of TBL closely mirror established clinical interventions for ADHD [19]. By decomposing complex, attention-demanding goals into short, clearly bounded subtasks with explicit transitions and iterative feedback, TBL compensates for the executive control gaps that characterize ADHD learners. This structural framework addresses their struggles with task initiation and sustained attention [10, 63, 85, 98, 117].

2.3 Digital Technologies and LLM-Powered Tools for ADHD Learners

Digital technologies for children with ADHD broadly target three aspects: sensing and assessment, daily routines, and engagement. Ubiquitous and wearable systems measure hyperactivity, screen for neuro-developmental risks, and infer attention or engagement [1, 3, 6, 28, 46, 47]; some others scaffold parent-child co-regulation, task tracking, routine management, and in situ support [75, 80, 99–102]; and gamified and speech-interactive tools aim to sustain participation [17, 69, 84]. However, these systems typically sense, scaffold, or motivate isolated tasks rather than orchestrate a complete learning process. Existing work does not specify how the sequencing, pacing, and adaptive feedback of expert-led interventions such as TBL can be encoded into an interactive system.

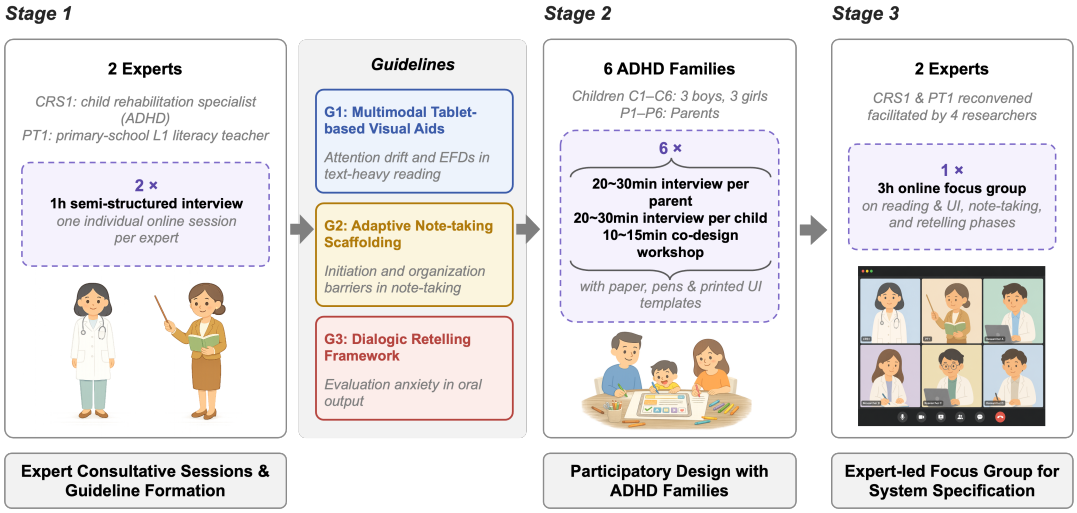


Fig. 2. An overview of our three-stage pilot formative study. Guidelines are organized top-down per reading, note-taking, and retelling components of the activities necessary for literacy learning.

Advances in Large Language Models (LLMs) offer a potential means of addressing this gap. Unlike traditional tutoring systems based on hard-coded rules [40], LLM-powered systems can sustain coherent dialogue and adapt to individual learner input [8, 48]. Educational applications have explored knowledge comprehension, teacher role-playing, and writing support [11, 67, 92, 114], while ubiquitous computing systems have grounded LLM interactions in sensed context or curated expert knowledge [74, 121, 122]. Yet these systems are largely designed for NT users and favor open-ended interaction over executive-function support. For children with ADHD, such interfaces may encourage cognitive offloading, increase choice overload, and impose executive demands that they are poorly equipped to manage [12, 43, 50, 58].

Although pedagogical and clinical frameworks such as TBL are well established in traditional settings [10], their translation into AI-driven systems remains underexplored. The resulting gap is twofold. First, there is no reference design pattern for translating a validated intervention structure into an LLM workflow while preserving its sequencing and scaffolding. Second, there is little interaction-level evidence on how children with ADHD appropriate such a digital learning route relative to their NT peers [82]. Addressing both gaps is necessary for designing neuro-inclusive interactive learning systems.

3 Pilot Formative Study with ADHD Families, Clinical and Educational Experts

To answer RQ1 and to design the probe system ReTell, we conducted a three-stage pilot formative study (see Figure 2). All collected data were anonymized and stored securely; the entire study, including a subsequent in situ study, received approval from the University Hospital's Institutional Review Board (IRB).

3.1 Stage 1: Expert Consultative Sessions and Guideline Formation

We conducted the 60-minute semi-structured online interview separately with a child rehabilitation specialist for ADHD (CRS1) and a primary school Chinese literacy (L1) teacher with extensive experience teaching students with ADHD (PT1). The experts were interviewed separately so that their agreement would serve as convergent evidence from independent sources rather than a consensus artifact of mutual influence [78]. Two researchers

independently open-coded the transcripts into pairs of expert-identified challenges and recommendations, clustered them thematically, and validated the resulting guidelines with each expert (see protocol and coding details in Appendix A).

Without knowing each other's input, the experts converged on two topics: *i) The ADHD-NT academic gap*: both experts described the well-documented gap between school-aged children with ADHD and their NT peers [7, 34]. CRS1 noted it usually emerges around Grades 3-4 when curricular demands sharpen; PT1 reported persistent underperformance in text-heavy reading, note-taking, and structured oral expression despite normal intelligence. This informed our participatory recruitment and study framing. *ii) A retelling-oriented TBL workflow*: both observed that dividing literacy learning into short, bounded subtasks with explicit transitions mirrors the compensatory strategies they often adopt in ADHD interventions; however, this well-practiced strategy did not extend well into the family scenarios; experts particularly identified retelling as a suitable output-type task, which resembles a "classroom-oriented method for motivating students' literacy comprehension" (PT1) and is consistent with "clinical intervention for structural oral expression" (CRS1). It should be noted that *retelling*, in this study, primarily refers to the retelling rehearsal organized by the system (i.e., the AI); each session then ends with the child retelling the article to their parents. Triangulating these perspectives, we adopted the retelling-oriented TBL workflow (reading with note-taking as input, retelling as output) to tackle the *challenge* and design the system:

(1) Guideline 1 (*attention drift and EFDs in text-heavy reading*): a multimodal approach with tablet-based visual aids to maintain attention and reduce demands on executive function;
(2) Guideline 2 (*initiation and organization barriers in note-taking*): adaptive note-taking scaffolding that lowers the threshold for organizing information, bridging comprehension and oral expression;
(3) Guideline 3 (*anxiety in oral expressions*): a dialogic framework to transform a rigid test into a natural interaction.

3.2 Stage 2: Participatory Design with ADHD Families

We recruited six families (children C1-C6, 3 boys and 3 girls, aged 10-12, Grades 4-6; parents P1-P6). All children had ADHD diagnoses confirmed within the past 6 months before the experiment, and no other conditions affecting reading, writing, or speaking existed. After we briefed each family on the study's goals, data recording, and the right to withdraw, parents signed written consent forms and children gave their assent. Each family completed two 20~30-minute semi-structured interviews, after which children participated in 10~15-minute co-design workshops to sketch UI elements (e.g., reading cards, AI avatars) and propose AI/UI assistance using printed templates (see Figure 14 and protocol details in Appendix A). Transcripts were thematically coded by two researchers [23]; sketches, annotated in situ with children's explanations, were cataloged and merged with the interview codes.

3.3 Stage 3: Expert-led Focus Group for System Specification

To complement family perspectives, we held a three-hour online focus group with CRS1 and PT1 to refine `ReTell` into a practical, accessible design. After reaching consensus on the retelling-anchored TBL workflow, the discussion covered (i) reading materials and UI for a tablet-based application, (ii) note-taking support, and (iii) the structure of retelling phases involving AI and parents. Four researchers facilitated the discussion; in situ notes were validated with experts during the session and refined against the transcripts.

3.4 From Findings to Design Goals

Figure 3 traces each guideline (G) and finding (F, reported as quotes from experts and participants) to the design goal (DG) and to the feature module (M) of `ReTell` that implements them accordingly. In detail, these include:

(1) **DG1: Facilitate reading comprehension and engagement** (Guideline 1). Participants confirmed that scaffolds are essential for maintaining attention and mitigating EFDs, helping "recognize complex Chinese

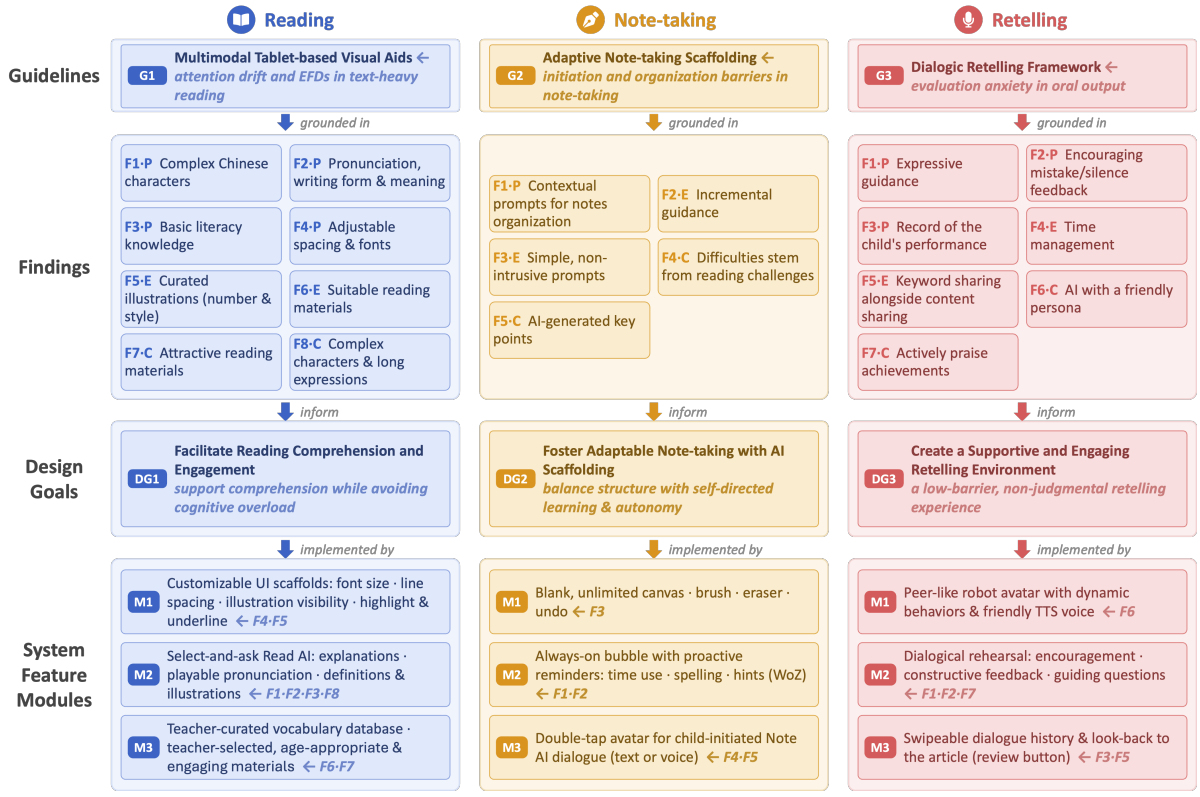


Fig. 3. Overview of how findings from our pilot formative study informed ReTell's design across reading (blue), note-taking (yellow), and retelling (red). The top layer presents the guidelines (**G**) distilled from expert consultative sessions, which guided the following stages. Building on these, findings (**F**) from parents (P), experts (E), and children with ADHD (C) motivate three design goals (**DG**), which are realized through three feature modules (**M**). IDs and colors are consistent across the system architecture and workflow figures.

characters" (P2) and explain "pronunciation, writing form, and meaning" (PT1, CRS1), with materials children "would be willing to read back home" (C1-C6), complemented by embedded literacy aids (e.g., visual adjustments, audio pronunciation, in-text definitions) (P2, P4, P6, CRS1), and curated, adjustable illustrations (PT1). In this regard, to support comprehension while avoiding cognitive overload, ReTell provides customizable UI elements (font size, line spacing, illustration visibility) and an on-demand, select-to-ask AI assistant backed by a curated vocabulary database.

(2) **DG2: Foster adaptable note-taking with AI scaffolding** (Guideline 2). Participants highlighted the challenge of note-taking for children with ADHD, with PT1 noting that "presenting these children with a blank space for note-taking is challenging"; participants wanted just-in-time contextual prompts (keywords, core points, structural hints) that are "simple enough to lower the barrier for children, yet not so intrusive" (CRS1). Accordingly, to balance structure with self-directed learning, ReTell settled on a blank, unlimited canvas instead of rigid templates, and its AI scaffolding combines proactive reminders (e.g., time spent, spelling errors, note-taking hints) with child-initiated dialogue, which accommodates EF challenges while promoting autonomy in note-taking.

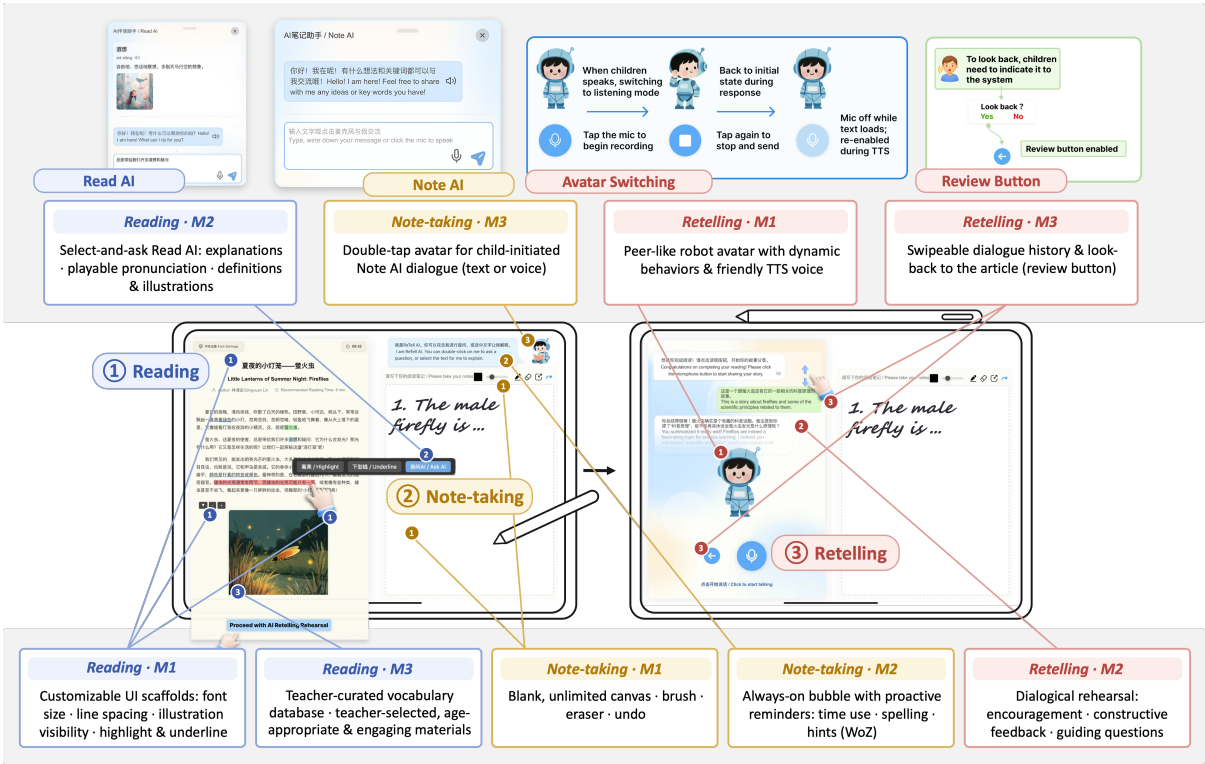


Fig. 4. User interface of ReTell across three phases: Reading (blue), Note-taking (yellow), and Retelling (red). Each callout describes a feature module (**M**), with circled numbers (1, 2, 3) indicating its location on the corresponding screen area. Module IDs and color coding follow Figure 3 of Section 3.

(3) **DG3: Create a supportive and engaging retelling environment** (Guideline 3). The dialogic framing was deemed to make the intervention feel like a natural activity rather than a test: children "liked to practice oral skills with an AI that simulates a peer and actively praises achievements" (C5), with expressive guidance (P3, P6) and encouraging feedback during mistakes or silences (P4). Therefore, ReTell offers a peer-like AI coach with dynamic avatar behaviors, constructive feedback and guiding questions, plus a visible dialogue history and look-back mechanism for a low-barrier, non-judgmental experience.

4 ReTell: an LLM-organized TBL Application

This section specifies how the system works and where the LLM is involved, which helps distinguish the roles of system components and human-in-the-loop elements in the observed patterns. Each use of AI is also logged individually (what was invoked, at which phase, and with what input and output).

4.1 The ReTell Framework

ReTell is a tablet-based, LLM-powered web application, built with a Vue 3, TypeScript, and Vite frontend and a Flask backend with SQLite. It divides the screen into two parallel panels: the left panel houses the reading and, later, retelling rehearsal modules; the right panel is a fixed note-taking area with a toolbar and canvas (Figure 4 shows the default right-handed layout). The GUI adopts a children-adapted style with a dyslexia-friendly yellow

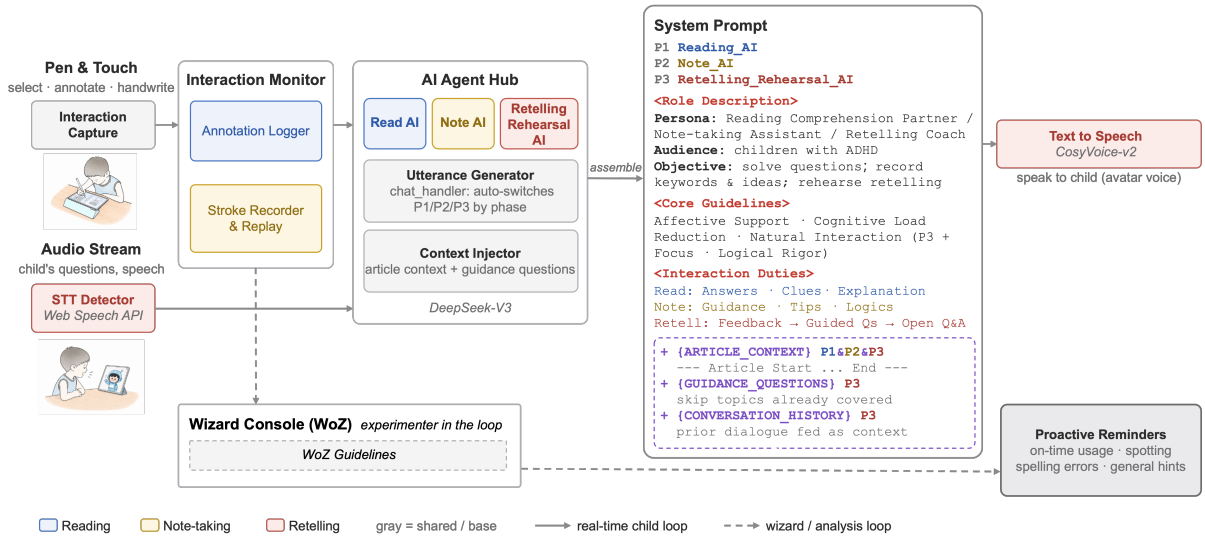


Fig. 5. System architecture and runtime workflow of ReTell. Children's pen-and-touch and speech inputs are recorded by the *Interaction Monitor*, coordinated through the *AI Agent Hub*, grounded with phase-specific context by the *Context Injector*, and returned as voice feedback. The *Wizard Console* supports experimenter-operated reminders and corrections.

background [87] and adjustable multimodal elements. Every feature module directly implements the findings of our formative study, i.e., Figure 3 and Figure 4 share the same set of module IDs (M1-M3 per phase) and colors, tracing each on-screen feature back to the findings and design goal it addresses. As shown in Figure 4, a full TBL session of ① to ③ includes:

(1) *Reading (knowledge input).* The child reads a teacher-selected article, optionally adjusting font size, line spacing, and illustration visibility, and marking the text with underlines or highlights. Selecting difficult text and tapping "Ask AI" opens a dialogue window where the *Read AI* returns a context-grounded explanation and, for words in the teacher-curated vocabulary database, playable pinyin, a child-friendly definition, and an illustration; the child may continue with typed or spoken follow-ups.
(2) *Note-taking (interleaved with reading).* The child handwrites freely on the unlimited canvas. Assistance comes via two pathways: an always-on bubble delivering proactive reminders (time use, spelling, hints), with a Wizard-of-Oz design (detailed in Appendix B.2); children can also double-tap the avatar to initiate a *Note AI* dialogue using text or voice input.
(3) *Retelling rehearsal (knowledge output).* The article area here is replaced with an avatar and a microphone button; the notes stay visible as a reference. The child can tap the microphone to start a dialogue turn, during which speech is transcribed in real time, and the *Retelling Rehearsal AI* replies with encouragement, feedback, or a guiding question, delivered via the avatar and accompanied by a rolling dialogue history. In this phase, the child can also tap "look back" to return to the article, and is allowed to loop freely between rehearsal and look-back. Children may also revise their notes during rehearsal.
(4) *Retelling to parents.* The child then retells the article to their parents, using their notes as support. This final step deliberately shifts from AI-assisted rehearsal to authentic interpersonal communication: the AI supports preparation, while the meaningful communicative act takes place with a real human audience.

4.2 System Architecture and LLM Integration

As shown in Figure 5, ReTell receives children's pen-and-touch interactions and speech input, with the latter transcribed through the Web Speech API. An *Interaction Monitor* timestamps interface adjustments, text markup, AI activations, and pen strokes in SQLite, supporting both real-time session operation and component-level usage logs for our in situ analysis. An *AI Agent Hub* coordinates three phase-specific agents, namely *Read AI*, *Note AI*, and *Retelling Rehearsal AI*, which share a *child-like humanoid robot* avatar. An Utterance Generator switches the avatar's persona across phases, as detailed in Appendix B. To ground each response, a *Context Injector* dynamically supplies the instant prompt with `ARTICLE_CONTEXT`, teacher-authored `GUIDANCE_QUESTIONS`, and, during retelling, `CONVERSATION_HISTORY`. DeepSeek-V3[1] generates the responses, which are spoken through CosyVoice-v2 text-to-speech (TTS) service[2]. Appendix B.2 further distinguishes the implementation provenance of each component, including deterministic UI scaffolds, teacher-curated content delivered through an LLM, fully LLM-generated dialogue, and WoZ-operated proactive reminders.

4.3 Embedding Probes behind ReTell's Function Modules

Stakeholder input left open how children with ADHD would engage with AI support. For example, parents preferred timed tasks (P2), whereas CRS1 emphasized autonomy and a balance between effectiveness and sustained attention. We therefore designed ReTell's modules as "probes" that pair optional support with behavioral logging. The subsequent in situ study could thus examine not only how children with ADHD and NT peers perceived each component, but also whether and how they used it:

(1) **Reading (DG1):** Separately logging non-LLM scaffolds and the activation of *Read AI* [57, 59] probes whether children rely on interface customization, AI explanations, or both, and how they balance assistance and autonomy.
(2) **Note-taking (DG2):** The blank canvas reveals spontaneous strategies and difficulties, while logs of wizard-triggered reminders and child-initiated double-tap dialogues probe responses to system-initiated versus self-directed support.
(3) **Retelling rehearsal (DG3):** Rehearsal turns, look-back requests, and history swipes provide behavioral traces of engagement with active, encouraging AI coaching and its potential to sustain oral-expression practice.

5 In Situ Study with Comparative Probe Designs

Targeting RQ2, we conducted an in situ study with both children with ADHD and their NT peers to distinguish their differences in interacting with ReTell and extract design implications.

5.1 Participants

As listed in Table 1, we recruited 43 families via flyers distributed through social media and local mall, school, and hospital publicity boards: 21 children with ADHD (A1-A21, parents PA1-PA21; age: M = 9.25, SD = 1.17; 16 boys and 5 girls, consistent with China's reported ADHD male-to-female ratio of approximately 2.6 [60]) and 22 NT children (N1-N22, parents PN1-PN22; age: M = 9.74, SD = 1.47; 11 boys and 11 girls). All children had completed grades 3-6 and received school-based literacy and narration training; each family received a 15 USD transportation allowance. ADHD subtypes are reported for completeness but not analyzed, as they were assessed with heterogeneous scales (i.e., SNAP-IV, DSM-IV/5) whose updates increasingly reflect preliminary symptom presentation, fluctuate over time, and fall outside our focus on overall ADHD-NT comparisons. Sessions took place in quiet rooms provided by five communities across the city (Figure 6) and were selected for the families' convenience; upon arrival, a researcher introduced the study again and obtained written informed consent.

[1]DeepSeek-V3-0324, DeepSeek, https://api-docs.deepseek.com/.
[2]CosyVoice 2 with its longjielidou_v2 tone, Alibaba, https://funaudiollm.github.io/cosyvoice2/.

Table 1. Table of Participants. A/N=ADHD/NT Children. We add the character 'P', e.g., PA1, to indicate the parent of children A1. The subtype refers to the result obtained from either SNAP-IV, DSM-IV, and/or DSM-5 reports provided by the family.

ID	Gender	Age	ADHD Subtype	ID	Gender	Age	Type
A1	Girl	9	Inattentive	N1	Girl	12	Neurotypical
A2	Boy	9	Inattentive	N2	Boy	9	Neurotypical
A3	Boy	10	Combined	N3	Boy	10	Neurotypical
A4	Boy	9	Inattentive	N4	Girl	11	Neurotypical
A5	Girl	11	Inattentive	N5	Girl	10	Neurotypical
A6	Boy	10	Inattentive	N6	Boy	9	Neurotypical
A7	Girl	9	Combined	N7	Girl	9	Neurotypical
A8	Boy	12	Inattentive	N8	Boy	9	Neurotypical
A9	Boy	10	Combined	N9	Boy	9	Neurotypical
A10	Boy	10	Combined	N10	Boy	9	Neurotypical
A11	Boy	9	Combined	N11	Girl	10	Neurotypical
A12	Boy	12	Inattentive	N12	Boy	11	Neurotypical
A13	Girl	9	Inattentive	N13	Boy	13	Neurotypical
A14	Boy	8	Combined	N14	Girl	13	Neurotypical
A15	Boy	8	Inattentive	N15	Boy	11	Neurotypical
A16	Boy	9	Combined	N16	Boy	12	Neurotypical
A17	Girl	7	Inattentive	N17	Girl	10	Neurotypical
A18	Boy	11	Combined	N18	Girl	8	Neurotypical
A19	Boy	10	Inattentive	N19	Girl	12	Neurotypical
A20	Girl	8	Inattentive	N20	Girl	12	Neurotypical
A21	Boy	8	Inattentive	N21	Girl	10	Neurotypical
				N22	Boy	12	Neurotypical

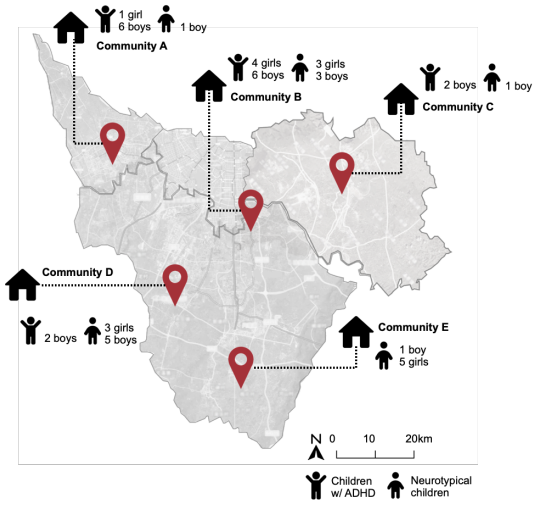


Fig. 6. The illustration of our five experimental sites in Community A, B, C, D, and E. The map is blurred, cut into pieces, and reorganized to maintain anonymity.

5.2 Parent-filled Pre-experiment Questionnaires as a Naturalistic Baseline

Because our comparative probe design did not include a controlled baseline condition, we used pre-experiment questionnaires (Appendix D) to establish a *naturalistic baseline* based on parents' reports of their children's reading, note-taking, and retelling practices at home before exposure to ReTell. These reports were not collected with the same tasks and outcomes as the ReTell sessions, so this baseline describes the groups' backgrounds rather than serving as a pretest for measuring change. It characterizes the two groups along two complementary dimensions. In terms of learning opportunity and technology exposure, Mann-Whitney U tests revealed no significant group differences in self-directed reading, parental reading, family-based writing training, narration behaviors, or experience with e-readers and AI devices. Given our sample size, however, these non-significant results cannot rule out smaller differences between the groups. In terms of literacy ability and engagement, these two groups did differ in the skills ReTell targets: parents reported that children with ADHD exhibited weaker comprehension during reading ($Q_{3.3}$, $p = 0.0029$), fewer note-taking habits during reading at home ($Q_{2.1}$, $p = 0.0089$), and less clear and organized oral expression ($Q_{3.2}$, $p = 0.0023$). This pattern is consistent with difficulties in executive function associated with ADHD in cognitively demanding tasks [18, 56], as both experts in our formative study also independently noted. We use this baseline as context when interpreting children's behaviors within the system throughout the findings. Persistent differences between groups are treated as likely reflecting characteristics that existed before the study, whereas departures from patterns reported at home are triangulated across quantitative and qualitative data as observations that may relate to ReTell's TBL scaffolding, without implying a causal effect.

5.3 Procedure & Materials

The experiment followed a five-stage sequential design:

(1) **Pre-experiment Preparation (10-15 min)**: Parents completed the pre-study questionnaire in the waiting room, while children watched an instructional video and read the warm-up article with the experimenter's guidance until ready.
(2) **`ReTell` with a Short Article (20-30 min)**: Children completed reading, note-taking, and retelling rehearsal with `ReTell`, after which parents joined for the final retelling session; the experimenter was absent at this stage.
(3) **Break Interval (5-10 min)**: A short break was included, following a child physiotherapist's recommendation.
(4) **`ReTell` with a Long Article (25-35 min):** Children repeated procedure 2 with a longer article.
(5) **Post-Experiment Interview (15-20 min):** A semi-structured family interview covered children's experiences with `ReTell` and parents' perceptions of usability and effectiveness.

Each session used the following equipment: a 12.9 inch iPad Pro with Apple Pencil; a DJI Mic Mini 2 for enhanced voice input; a laptop for playing the instructional video upon arrival; a laptop running `ReTell`'s backend, and another laptop livestreaming child performance to parents; two extra laptops for the experimenter's observational notes and interview recording; and a smartphone recording video of the entire session. Post-study interviews involved three researchers: a primary interviewer, a notetaker, and a third researcher who identified gaps or issues requiring clarification. All interview data were audio-recorded and transcribed for analysis. The reading materials and their teacher-reviewed illustrations are detailed in Appendix C.

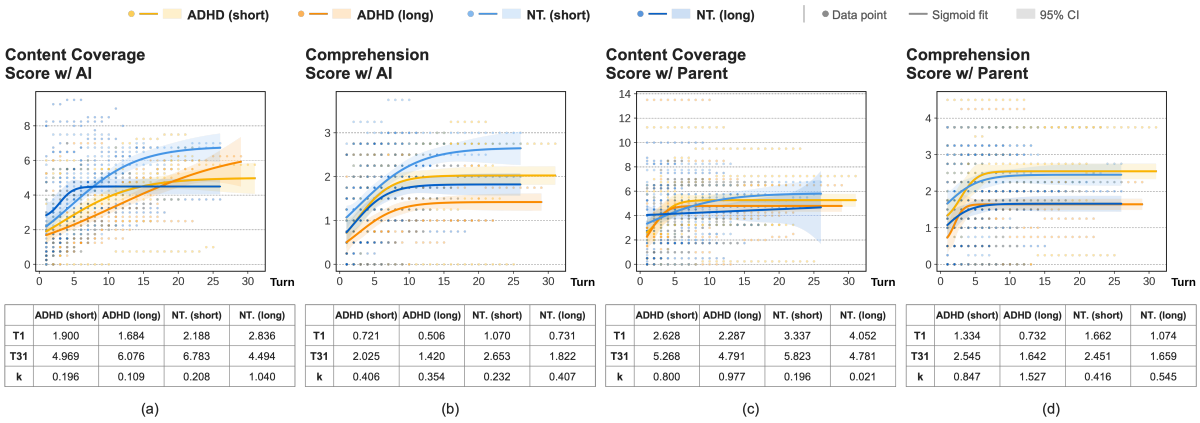
5.4 Data Collection & Analysis

We adopted a mixed-methods approach that combined on-site observations, system logs, expert-rated performance, and interviews to uncover interaction patterns, challenges, and needs of children with ADHD when using an AI TBL application, compared with their NT peers.

5.4.1 Annotation of Children's Behaviors. A behavioral coding framework was developed with clinical and educational experts, categorizing objective behaviors into **(1) temporal features** (e.g., time on reading or note-taking), **(2) interaction patterns** (e.g., UI adjustments, types of AI queries), and **(3) bodily behaviors** (e.g., mistaken touches, signs of attention loss). Two researchers independently coded 10% of the data to align the rubrics before coding the full dataset, with a third researcher cross-checking the annotations. For task performance, two primary school teachers rated the *relevance* and *richness* of notes on 5-point Likert scales and classified note forms (e.g., doodles vs. mindmaps) under standards unified through prior calibration; for rehearsal/retelling, researchers coded key-point coverage and answer correctness from speech transcripts against teacher-provided references, double-checked by the teachers. Details of the framework are shown in Supp. Table 1.

5.4.2 Quantitative Analysis. As annotated behaviors and retelling performances violated normality and sphericity assumptions (Shapiro-Wilk and Mauchly's tests), we applied non-parametric methods: Mann-Whitney U (MWU) tests for between-group comparisons and Wilcoxon Signed-rank (WS) tests for within-group comparisons (e.g., short vs. long articles), with Bonferroni corrections for multiple comparisons (corrected $\alpha = 0.05/4 = 0.0125$). Comparison results that did not meet this threshold, as well as counts and proportions that were not statistically tested, are reported descriptively without inferential interpretation.

5.4.3 Qualitative Analysis. Post-experiment interviews were analyzed via thematic analysis [23] under four categories aligned with our DGs: (1) children's reading experiences, (2) note-taking behaviors and preferences, (3) retelling experiences, and (4) parents' perspectives. Two researchers who organized the interviews independently



	ADHD (short)	ADHD (long)	NT. (short)	NT. (long)
T1	1.900	1.684	2.188	2.836
T31	4.969	6.076	6.783	4.494
k	0.196	0.109	0.208	1.040

(a)

	ADHD (short)	ADHD (long)	NT. (short)	NT. (long)
T1	0.721	0.506	1.070	0.731
T31	2.025	1.420	2.653	1.822
k	0.406	0.354	0.232	0.407

(b)

	ADHD (short)	ADHD (long)	NT. (short)	NT. (long)
T1	2.628	2.287	3.337	4.052
T31	5.268	4.791	5.823	4.781
k	0.800	0.977	0.196	0.021

(c)

	ADHD (short)	ADHD (long)	NT. (short)	NT. (long)
T1	1.334	0.732	1.662	1.074
T31	2.545	1.642	2.451	1.659
k	0.847	1.527	0.416	0.545

(d)

Fig. 7. Sigmoid regression curves of content coverage and reading comprehension by turns for (a–b) children's rehearsal with AI and (c–d) retelling with parents. T1 = calculated value at turn 1 (lower asymptote), T31 = calculated value at turn 31 (upper asymptote, defined by maximum turns across all participants), K = curve steepness (growth rate of scores with turns).

coded transcripts and field notes, refined the codebook through discussion, and a third researcher reviewed all coded data to resolve remaining disagreements.

6 Results

In this section, we integrate quantitative and qualitative findings from the in situ study to address RQ2. Children's in-system behaviors are interpreted against their *naturalistic baseline*. Because ReTell was experienced as an integrated system, the reported outcomes characterize performance under its LLM-organized TBL workflow rather than the effect of any individual component; component provenance is noted only where relevant (Appendix B.2). Section 6.1 examines in-session retelling performance, Section 6.2 analyzes children's interaction patterns and parental support, and Section 6.3 derives design implications from these findings. Distraction-related behaviors are reported in Appendix E.

6.1 The Effect of LLM-organized TBL on Children's Retelling Performance

6.1.1 Neurotypical Children Started Higher, while Children with ADHD Improved over Additional Interaction Turns. To characterize children's in-session outcomes during LLM-led rehearsal and parent-led retelling, we analyzed their entire conversations. Sigmoid regression was applied to content coverage and reading comprehension computed for each dialogic turn across all participants in the same group (see Figure 7). In LLM-led rehearsal (Retelling M2), NT children often started with higher scores on both metrics, consistent with the stronger baseline literacy skills reported in the pre-experiment questionnaires ($Q_{3.2}$, $Q_{3.3}$). As the rehearsal progressed, children with ADHD gained steadily over additional turns of interaction with the *Retelling Rehearsal AI*. During the long-article session, NT children showed signs of fatigue and impatience, frequently stopped the rehearsal early, which produced a steeper fitted curve ($k = 1.040$, Figure 7(a)), whereas some children with ADHD who rehearsed longer reached higher fitted final coverage ($\mathbf{ADHD}_{\mathbf{T31}} = 6.076$ vs. $\mathbf{NT}_{\mathbf{T31}} = 4.494$). Overall, no statistically significant group differences were detected in coverage or comprehension during rehearsal with the AI. In parent-involved retelling, NT children again showed higher initial performance, while children with ADHD exhibited steeper fitted growth ($ADHD_k > NT_k$) over more interaction turns and parental guidance, and by the end of the session,

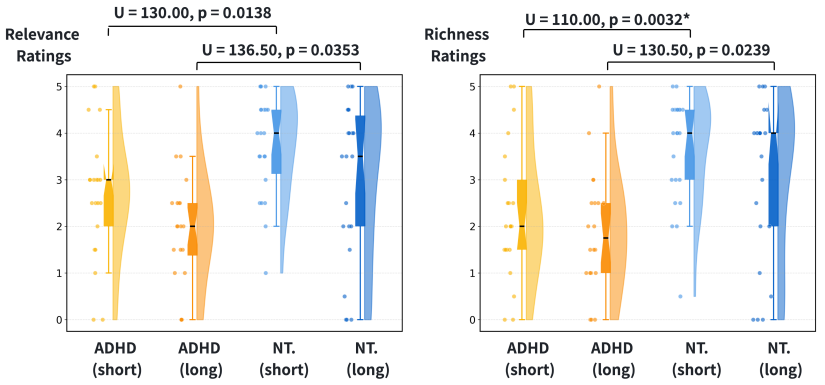
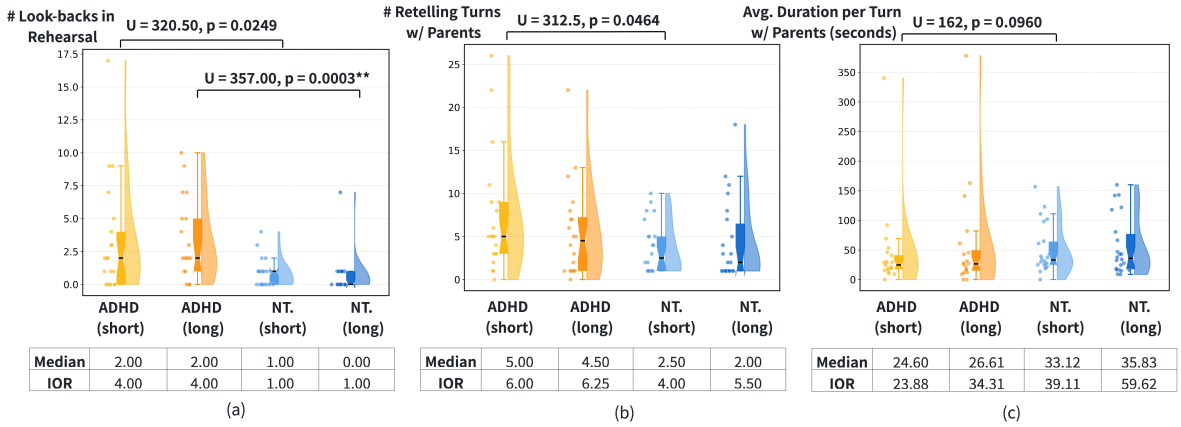


Fig. 8. Boxplots of experts' ratings of children's notes for (a) relevance and (b) richness, comparing ADHD and NT groups. A minor random jitter (0.01–0.09) was added for visualizing discrete ratings without affecting statistical analysis.

no statistically significant group differences were detected in coverage or comprehension. The absence of a significant difference, however, does not by itself indicate equivalent performance.

Expert ratings of children's notes (Figure 8) showed a significant group difference in note richness in the short-article session ($p = 0.0032 < 0.0125*$), with NT children rated higher. Descriptively, NT children received higher richness ratings in the long-article session and higher relevance ratings, whereas children with ADHD produced more doodling and irrelevant content during the long-article session. However, none of these differences met the corrected significance threshold (e.g., richness: $p = 0.0239$; doodling: $\mathbf{U} = 290.00$, $p = 0.0321$). Note structure varied substantially across individuals in both groups, from unstructured keywords to color-coding and mind maps (examples shown in Supp. Figure 1). These differences align with the pre-existing gap in note-taking capability between the two groups, as reported in the pre-experiment questionnaires ($Q_{2.1}$). Despite these differences in note taking, retelling outcomes showed no statistically significant group differences, suggesting that children with ADHD may have drawn on other resources during the task, which we report in Section 6.1.2.

6.1.2 Look Backs as a Possible Compensatory Strategy for Children with ADHD during Rehearsal. The clearest interaction pattern observed alongside these outcomes was children's revisiting of the article through Retelling M3, which may represent a possible compensatory strategy: children with ADHD invoked it significantly more often during the long-article rehearsal ($\mathbf{U} = 357$, $p = 0.0003 < 0.0125 * *$) (see Figure 9(a)); look-backs were numerically more frequent among children with ADHD in the short-article session as well, though this difference did not reach the corrected threshold ($\mathbf{U} = 320.5$, $p = 0.0249$). Children attributed the look-backs to the rehearsal dialogue itself: *"AI reminds me about contents that I am unfamiliar with"* (A14, A3) and *"I understood the article better (at second visit) given my interactions with AI"* (A10); parents similarly emphasized that the *Retelling Rehearsal AI "organized the thoughts (of my child) more clearly"* and *"helped (the child) cover more details of the article"* (PA6). Regarding conversational pacing, turn counts were numerically higher and per-turn durations numerically shorter among children with ADHD across both phases, but none of these comparisons reached the corrected threshold (long-article rehearsal turns: $p = 0.0520$; short-article retelling turns: $p = 0.0464$; per-turn durations: $p = 0.0960$; see Figure 9). We noticed that some children revised or added notes during look-backs, but these behaviors showed no significant group differences.



	ADHD (short)	ADHD (long)	NT. (short)	NT. (long)
Median	2.00	2.00	1.00	0.00
IOR	4.00	4.00	1.00	1.00

(a)

	ADHD (short)	ADHD (long)	NT. (short)	NT. (long)
Median	5.00	4.50	2.50	2.00
IOR	6.00	6.25	4.00	5.50

(b)

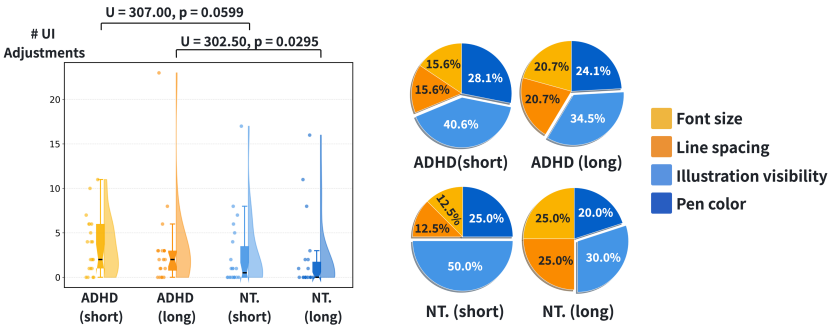
	ADHD (short)	ADHD (long)	NT. (short)	NT. (long)
Median	24.60	26.61	33.12	35.83
IOR	23.88	34.31	39.11	59.62

(c)

Fig. 9. Boxplots showing (a) number of times children look back at the article during AI-led rehearsal sessions, (b) number of dialogue turns, and (c) average duration per turn during the parent–child retelling phase.

6.2 Revealing ADHD-Specific Interaction Patterns with ReTell

6.2.1 Adaptive Patterns in Active UI Customization. Regarding the customizable UI scaffolds (Reading M1), we did not detect significant group differences in adjustment frequency in either the short- or the long-article task ($p = 0.0295$, long-article; Figure 10); descriptively, adjustment frequencies were numerically higher among children with ADHD. Both groups increased line-spacing adjustments for the longer article, explaining that wider spacing *"makes my reading easier"* (A2) and prevents *"repeatedly reading the same line"* (A16). Adjusting illustration visibility was the most frequent action with mixed motivations: closing illustrations to avoid distraction, enlarging them to grasp the theme or details (A4, A5), or shrinking them for easier navigation once the details were known (A7, A14, N8, N9). During parent-involved retelling, children employed four UI-supported reference strategies: "Retelling Following Notes" was more frequent among NT children (13 vs. 7 in long-article tasks; 14 vs. 11 in



	ADHD (short)	ADHD (long)	NT. (short)	NT. (long)
Median	2.00	2.00	0.50	0.00
IOR	5.00	2.25	3.50	1.75

(a) (b)

Fig. 10. (a) Boxplots and pie charts summarizing UI adjustment frequencies per participant during reading and note-taking of ADHD and NT children. (b) Pie charts showing the distribution of UI adjustment types across the two groups.

short-article tasks); children with ADHD ($N = 9$) relied more on "Retelling Following Original Text"; "Retelling Following AI Rehearsal Interaction History" was used less overall (NT $N = 6$ slightly higher in long-article tasks); and "Direct Retelling without notes" was similarly frequent across groups, used by nearly half of the children.

6.2.2 Frictions: Note-taking Challenges and Low Discoverability of AI Features. During the knowledge-input phase, children with ADHD spent numerically less time on reading and note-taking in the long-article session (Median = 324.50, IQR = 299.25) than NT peers (Median = 646.0, IQR = 503.0), but this difference did not reach the corrected threshold ($\mathbf{U} = 124$, $p = 0.0162$). In the short-article session, NT children produced significantly more notes ($p = 0.0063 < 0.0125*$), consistent with their stronger note-taking habits at home ($Q_{2.1}$). However, their note-taking declined significantly in the long-article session ($p = 0.0052 < 0.0125*$), suggesting possible task fatigue or reduced engagement after the initial freshness. Children with ADHD maintained a consistently low note-taking frequency across sessions and produced significantly longer strokes in the long-article session ($p = 0.0028 < 0.0125*$), with inspection revealing more doodling and irrelevant content (Figure 12(a)(b)) (Note-taking M1). The highlighting tool (Reading M1) was rarely adopted (six ADHD and five NT children) and described as *"not very useful"* (N1) or *"unfamiliar and inconvenient"* (A3).

Overall usage of *Read AI* (Reading M2) and *Note AI* (Note-taking M2/M3) was similarly low (Figure 11). In post-experimental interviews, six children with ADHD reported either *"forgetting the availability of the function"* (A1) or being *"uncertain about how to properly activate them"* (A11). Although some recognized potential benefits, e.g., *"Next time I want to use it, I believe it could help me"* (A4), they often missed opportunities to use it. Among actual usage, about half of the children with ADHD engaged with *Read AI* in both sessions, compared with fewer NT peers. The primary purpose was sentence-level comprehension, followed by word lookups. In contrast, *Note AI* was used only sporadically, with just two children with ADHD requesting note-specific assistance that resembled *Read AI*-like queries. Proactive support faced a different friction: children often ignored the WoZ reminders when they considered the flagged issues unimportant (N2, A8).

Both groups interrupted the *Retelling Rehearsal AI* (Retelling M1/M2), often cutting off its ongoing TTS output. Descriptively, interruptions were numerically more frequent among children with ADHD in the short-article session, and NT children's interruptions numerically increased from the short- to the long-article session; neither comparison reached the corrected threshold ($\mathbf{U} = 294$, $p = 0.0474$; within-group $p = 0.0320$; Figure 12(c)). The

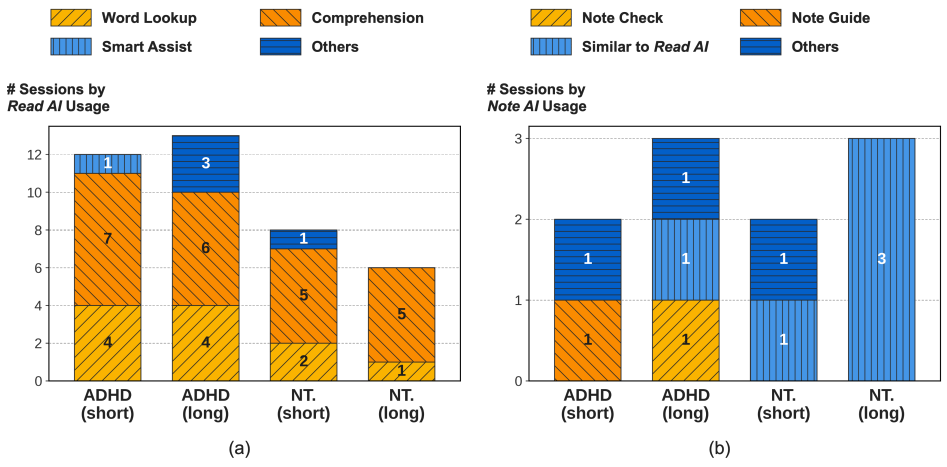


Fig. 11. The stacked histograms of the number of short/long article sessions where (a) *Read AI* and (b) *Note AI* were used.

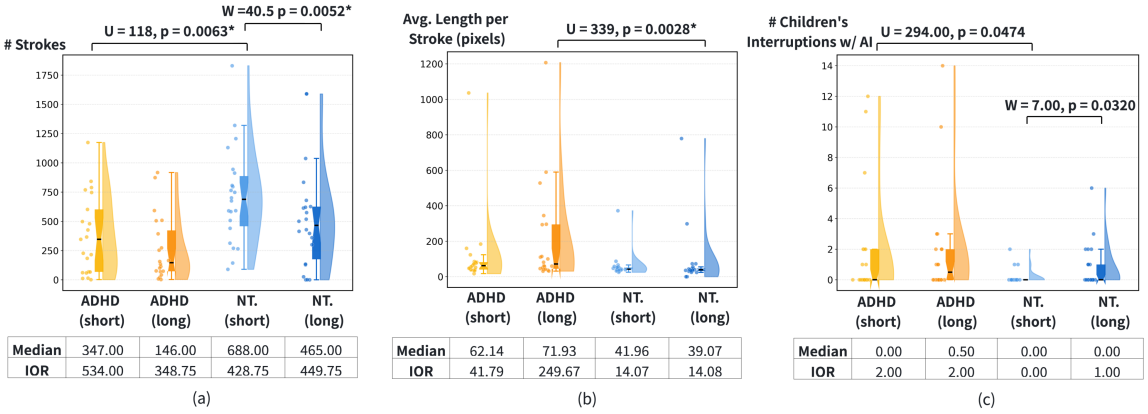


Fig. 12. The boxplots of note-taking behaviors: (a) the number of strokes and (b) average writing length per stroke, and rehearsal behaviors: (c) the number of interruptions with AI presented by children. NT.=Neurotypical.

interview data consistently identified two distinct friction sources behind the interruptions: children with ADHD described the AI's responses as *"repetitive"* (A16) or *"unhelpful"* (A15), while NT children cited unstable speech recognition, *"speech recognition was unstable, so I needed to pause and speak again"* (N8).

6.2.3 Parental Interaction Styles during the Retelling Phase. Parental behaviors during the AI-free retelling phase fell into two categories (Figure 13): *passive response*, (i.e., minimal reactions such as nodding or short utterances) and *proactive guidance*, (i.e., prompting questions when the child paused or directly steering the conversation, sometimes with signs of impatience). Parents from ADHD families showed proactive guidance more frequently than those from NT families across both tasks. Whether such early intervention limits children's independent narrative construction cannot be established from our observational data; we flag it as a candidate tension for future investigation rather than a finding.

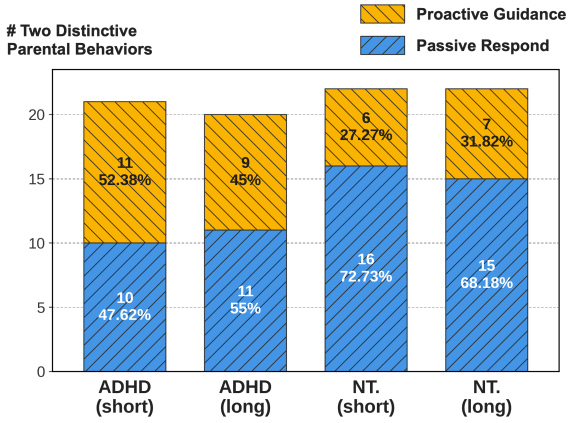


Fig. 13. Stacked histograms of parental behaviors during the retelling phase, showing counts of passive responses vs. proactive guidance in short- and long-article sessions. Each parent was classified into a single interaction pattern per session.

6.3 Design Implications

We distill four design implications for future interactive and ubiquitous learning systems that target similar populations. Each is anchored to the specific finding reported above, and we explicitly mark where an implication extends beyond what our data can directly support.

6.3.1 Scaffolding the Compensatory Needs: Adaptive Pacing and External Memory. The clearest compensatory pattern involved the look-back function: children with ADHD used it significantly more often during long-article rehearsal, while children and parents described AI-prompted revisits as supporting comprehension (A14, A3, A10, PA6). Children with ADHD also showed a descriptive tendency toward more frequent but shorter conversational turns, consistent with clinical recommendations for task segmentation and immediate feedback [29]. These findings suggest two requirements for system design. First, interaction history can serve as an external memory representation [93]: persistent traces, such as our sliding dialogue history and look-back mechanism, help learners reconnect information across temporal gaps rather than relying solely on working memory [68, 86]. Second, when the goal is comprehension or task completion rather than processing speed or fluency, session logic should avoid imposing uniform time limits. Adaptive pacing [24] should accommodate slower knowledge construction that does not follow a linear path.

6.3.2 Reducing Interaction Cost through Mixed-Initiative, Discoverable Affordances. The passive assistants (*Read AI* and *Note AI*) were used infrequently and sporadically, despite children recognizing their value (A4). Their accounts pointed to activation barriers: some forgot that the functions were available (A1), while others were unsure how to invoke them (A11). This illustrates an execution activation barrier in which users recognize the value of a function but cannot readily translate their intention into the required interface action [123, 124]. Proactivity alone, however, did not ensure engagement, as children often ignored the WoZ reminders (see Section 6.2.2) [70]. Together, these findings suggest that neither purely user-initiated nor uniformly proactive assistance is sufficient. In this regard, a mixed-initiative design [4, 41, 62] should make assistance visible and easy to activate, while also offering support proactively when interaction cues indicate that a child may need help, such as repeated reading or prolonged inactivity [51]. This approach avoids both silent assistance, which depends entirely on recall, and fixed-schedule interruptions, which may lack contextual relevance.

6.3.3 Lowering the Cost of Externalization with Multimodal Input. Children with ADHD produced significantly longer pen strokes and more doodling during the long-article session, and their notes received lower richness ratings in the short-article session. Their feedback suggests that the input method may have contributed to these difficulties: one child reported that *"I found it a bit difficult to write with the stylus"* (A13), while others explicitly requested voice input (A15, A18). The highlighting tool was also rarely used and was described as *"not very useful"* (N1) or *"unfamiliar and inconvenient"* (A3). These findings suggest that note-taking support should prioritize low-effort externalization rather than enforcing a conventional note format, for example, through voice input and AI-assisted organization [66, 90]. The rehearsal dialogue history, which several children already consulted during retelling, could likewise be developed into a structured alternative or complement to handwritten notes.

6.3.4 Repositioning AI within the Child-AI-Parent Triad. Descriptively, the AI addressed a larger share of children's factual errors than parents did (57.7% vs. 17.0%; see examples in Supp. Figures 2 and 3), and parents in ADHD families more often exhibited proactive guidance, sometimes with signs of impatience, consistent with documented parenting stress in ADHD families [72, 108]. These observations suggest a share of labor: offloading repetitive factual correction to the AI so that parents can redirect limited resources toward emotional validation and narrative scaffolding [36, 95, 97]. We frame the further role of AI as an emotional buffer that de-escalates parent-child tension [20, 35, 94] as a design hypothesis, motivated by but not validated in our single-session data. Nevertheless, a full evaluation of this role requires longitudinal deployment in homes [13, 38].

7 Discussion

Here, we discuss what our observations imply for extending learning scaffolds beyond the screen and position LLM orchestration as a pathway to scaling expert-validated interventions. Finally, we acknowledge and delineate the limitations of our study and outline future directions in Section 7.3.

7.1 From Attention-Aware Constraints to Sensing-Driven Adaptivity for Children with ADHD

The LLM-organized TBL application was designed to sustain attention by breaking down demanding activities into short, bounded cycles, although our study did not directly measure attention. However, several components intended to provide support, including adjustable UI scaffolds, webpage interface artifacts, and peripheral devices such as the stylus and microphone, triggered documented episodes of distraction (see Appendix E). Notably, these episodes arose not from the instructional content of the scaffolds, but from their interactional affordances: browser controls, zooming interactions, and manipulable physical accessories all competed for children's attention. We derive a simple yet practical design principle to drive future developments, namely *attention-aware constraints*, for e-learning systems for children with ADHD. Rather than only sensing attention [5], systems can protect it by reducing nonessential affordances: i) separating presentation settings (e.g., font size) from learning sessions or at least de-emphasizing them during the session [52, 54, 65]; ii) disabling distracting gestures during reading, particularly given children's still-developing fine-motor control [42, 112]; and iii) exposing tools (e.g., the activation of stylus) only when they are relevant to the current activity. These lightweight constraints could reduce competing interactions and may help keep attention on the learning task [53], a possibility that future work should test.

Our findings (see Section 6.2.1, Section 6.3.3, and Appendix E) further suggest that distraction can arise not only from instructional content, but also from the visual and physical affordances surrounding it. This observation motivates exploration of voice-first, wearable, and ambient learning companions that reduce reliance on visually dense and physically manipulable interfaces [22, 37, 76, 107]. Such systems could also incorporate multimodal signals, including physiological and gaze data, to estimate attention-related states such as engagement and attentional focus [1, 6, 28, 47]. The resulting constraints could then adapt to the child's momentary state rather than remaining fixed throughout the session. Designing and validating these sensing-augmented off-screen systems remain important directions for future work.

7.2 Toward Bringing Expert-Led Interventions into Everyday Use through LLM Orchestration

The general takeaway from this paper concerns how LLMs may help operationalize expert-guided neuro-inclusive interventions. Domain experts, such as rehabilitation specialists and teachers, define and validate the intervention structure, while ReTell instantiates that structure in sessions held outside expert settings (see Section 5.1) by managing phase transitions, pacing activities, and generating multimodal scaffolds in response to children's input. Parents serve as the audience for children's final retelling (see Section 6.2.3). This architecture redistributes rather than replaces expert effort: experts contribute upstream through guidelines, curated materials, and workflow validation, while the system delivers that encoded expertise within each session. Although our evidence is limited to performance within single sessions (see Section 6.1), the findings suggest a possible design principle: begin with validated interventions and use LLMs to deliver them while aiming to preserve their structure, a practice similar to that in [104]; whether such delivery maintains structural fidelity remains to be evaluated. Future systems could extend orchestration from sequencing activities to timing support. In our study, scripted reminders were sometimes ignored, whereas passive assistance could be forgotten (see Section 6.2.2) [77]. A mixed-initiative system could instead use interaction signals, such as prolonged inactivity, rereading, or stalled turns, to determine when support is warranted [21, 41, 79]. Methodologically, our formative and in situ studies offer context-rich interaction patterns from a population for which data remain scarce [103]. Prior ubiquitous

systems have supported children with dyslexia or autism spectrum disorder (ASD) [94, 115, 125], yet it remains to be tested whether our traceable design process transfers to these neurodivergent populations.

7.3 Limitations and Future Work

Three limitations bound our claims. First, the study included no controlled comparison condition, as all participants used the same integrated system. The questionnaires completed before the experiment captured parents' perceptions of their children's habits and abilities; they were not an objective baseline measured with the same tasks and outcomes as the `ReTell` sessions. Consequently, we cannot determine whether `ReTell` caused the observed patterns or narrowed any existing gap between children with ADHD and NT children, and the patterns cannot be attributed to `ReTell` as a whole, let alone to its LLM component. Moreover, the absence of a statistically significant group difference does not, by itself, demonstrate equivalent performance; such a conclusion would require equivalence testing with a larger sample. Our per-component provenance delineation (Appendix B.2) makes component-level analysis possible in principle, yet isolating the contribution of LLM-generated dialogue will require future ablation studies, for example, by comparing the full system against a version with scripted dialogue. Second, all measures capture performance and behavior during a single visit: we did not assess retention or longer-term learning gains, and our findings concern performance within a single session only. Third, the small sample constrains statistical power; several trends are reported descriptively. Speech recognition errors in children's voices and occasional LLM misinterpretations further underscore that AI should complement, rather than replace, human support in developmentally consequential settings. In general, we see three directions: (i) longitudinal deployments of similar systems in homes, schools, and clinics with controlled comparisons to assess sustained engagement and retention; (ii) integrating multimodal sensing and advanced language model schemes for attention-adaptive, more reliable interaction; and (iii) extending the orchestration approach to other neurodiverse populations and to the wider support network, sharing progress with teachers and scaffolding parent-child joint activities so the AI becomes connective infrastructure rather than a standalone companion.

8 Conclusion

This paper presented the design, implementation, and in situ evaluation of `ReTell`, an interactive mobile system that operationalizes expert-validated TBL principles through a structured, LLM-organized workflow for children with ADHD. By restructuring otherwise open-ended AI interactions into context-grounded, phased tasks, `ReTell` aims to reduce the attentional and organizational demands that often disadvantage children with ADHD in independent learning. Our in situ study with 43 families (21 ADHD, 22 NT) detected no statistically significant group differences in retelling performance during the sessions, despite weaker literacy habits reported by parents of children with ADHD. Children with ADHD more often revisited the article and took shorter, more frequent dialogue turns, patterns that may reflect compensatory strategies worth supporting in ubiquitous AI learning systems. From these interaction-level findings, we derived design implications for neuro-inclusive interactive and ubiquitous systems, from treating interaction history as external memory to attention-aware constraints that motivate moving scaffolds beyond the screen. We hope this work informs future systems that treat cognitive diversity not as a deficit to be accommodated but as a design constraint, one that may also inform learning tools that serve a broader range of learners.

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References

[1] Yomna Abdelrahman, Anam Ahmad Khan, Joshua Newn, Eduardo Velloso, Sherine Ashraf Safwat, James Bailey, Andreas Bulling, Frank Vetere, and Albrecht Schmidt. 2019. Classifying attention types with thermal imaging and eye tracking. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 3 (2019), 1–27.

[2] Howard Abikoff, Richard Gallagher, Karen C Wells, Desiree W Murray, Lei Huang, Feihan Lu, and Eva Petkova. 2013. Remediating organizational functioning in children with ADHD: immediate and long-term effects from a randomized controlled trial. *Journal of consulting and clinical psychology* 81, 1 (2013), 113.

[3] Karan Ahuja, Dohyun Kim, Franceska Xhakaj, Virag Varga, Anne Xie, Stanley Zhang, Jay Eric Townsend, Chris Harrison, Amy Ogan, and Yuvraj Agarwal. 2019. EduSense: Practical classroom sensing at Scale. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 3 (2019), 1–26.

[4] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N Bennett, Kori Inkpen, et al. 2019. Guidelines for human-AI interaction. In *Proceedings of the 2019 chi conference on human factors in computing systems*. 1–13.

[5] Christoph Anderson, Isabel Hübener, Ann-Kathrin Seipp, Sandra Ohly, Klaus David, and Veljko Pejovic. 2018. A survey of attention management systems in ubiquitous computing environments. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 2 (2018), 1–27.

[6] Riku Arakawa, Karan Ahuja, Kristie Mak, Gwendolyn Thompson, Sam Shaaban, Oliver Lindhiem, and Mayank Goel. 2023. Lemurdx: using unconstrained passive sensing for an objective measurement of hyperactivity in children with no parent input. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 2 (2023), 1–23.

[7] L Eugene Arnold, Paul Hodgkins, Jennifer Kahle, Manisha Madhoo, and Geoff Kewley. 2020. Long-term outcomes of ADHD: academic achievement and performance. *Journal of attention disorders* 24, 1 (2020), 73–85.

[8] David Baidoo-Anu and Leticia Owusu Ansah. 2023. Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. *Journal of AI* 7, 1 (2023), 52–62.

[9] Russell A Barkley. 1997. Behavioral inhibition, sustained attention, and executive functions: constructing a unifying theory of ADHD. *Psychological bulletin* 121, 1 (1997), 65.

[10] Russell A Barkley. 2012. *Executive functions: What they are, how they work, and why they evolved.* Guilford Press.

[11] Jessie S Barrot. 2023. Using ChatGPT for second language writing: Pitfalls and potentials. *Assessing writing* 57 (2023), 100745.

[12] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman. 2025. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences* 122, 26 (2025), e2422633122.

[13] Erin Beneteau, Yini Guan, Olivia K Richards, Mingrui Ray Zhang, Julie A Kientz, Jason Yip, and Alexis Hiniker. 2020. Assumptions checked: How families learn about and use the echo dot. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 1 (2020), 1–23.

[14] Santiago Berrezueta-Guzman, Mohanad Kandil, María-Luisa Martín-Ruiz, Iván Pau de la Cruz, and Stephan Krusche. 2024. Future of ADHD care: evaluating the efficacy of ChatGPT in therapy enhancement. In *Healthcare*, Vol. 12. MDPI, 683.

[15] Joseph Biederman, Michael C Monuteaux, Alysa E Doyle, Larry J Seidman, Timothy E Wilens, Frances Ferrero, Christie L Morgan, and Stephen V Faraone. 2004. Impact of executive function deficits and attention-deficit/hyperactivity disorder (ADHD) on academic outcomes in children. *Journal of consulting and clinical psychology* 72, 5 (2004), 757.

[16] Aida Bikic, Brian Reichow, Spencer A McCauley, Karim Ibrahim, and Denis G Sukhodolsky. 2017. Meta-analysis of organizational skills interventions for children and adolescents with Attention-Deficit/Hyperactivity Disorder. *Clinical psychology review* 52 (2017), 108–123.

[17] Sabeel Butt, Fazal E Hannan, Mujahid Rafiq, Ibrar Hussain, CM Nadeem Faisal, and Waleed Younas. 2020. Say-It & Learn: Interactive Application for Children with ADHD. In *International Conference on Human-Computer Interaction*. Springer, 213–223.

[18] Anil Chacko, Brittany M Merrill, Michael J Kofler, and Gregory A Fabiano. 2024. Improving the efficacy and effectiveness of evidence-based psychosocial interventions for attention-deficit/hyperactivity disorder (ADHD) in children and adolescents. *Translational Psychiatry* 14, 1 (2024), 244.

[19] Hang Chan. 2023. Ideating task-based designs for special education learners. *Tesol Journal* 14, 3 (2023), e711.

[20] Huili Chen, Yubin Kim, Kejia Patterson, Cynthia Breazeal, and Hae Won Park. 2025. Social robots as conversational catalysts: Enhancing long-term human-human interaction at home. *Science Robotics* 10, 100 (2025), eadk3307.

[21] Woohyeok Choi, Sangkeun Park, Duyeon Kim, Youn-kyung Lim, and Uichin Lee. 2019. Multi-stage receptivity model for mobile just-in-time health intervention. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 3, 2 (2019), 1–26.

[22] Franceli L Cibrian, Kimberley D Lakes, Arya Tavakoulnia, Kayla Guzman, Sabrina Schuck, and Gillian R Hayes. 2020. Supporting self-regulation of children with ADHD using wearables: tensions and design challenges. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–13.

[23] Victoria Clarke and Virginia Braun. 2017. Thematic analysis. *The journal of positive psychology* 12, 3 (2017), 297–298.

[24] Albert T Corbett and John R Anderson. 2001. Locus of feedback control in computer-based tutoring: Impact on learning rate, achievement and attitudes. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 245–252.
[25] Samuele Cortese, Minjin Song, Luis C Farhat, Dong Keon Yon, Seung Won Lee, Min Seo Kim, Seoyeon Park, Jae Won Oh, San Lee, Keun-Ah Cheon, et al. 2023. Incidence, prevalence, and global burden of ADHD from 1990 to 2019 across 204 countries: data, with critical re-analysis, from the Global Burden of Disease study. *Molecular psychiatry* 28, 11 (2023), 4823–4830.
[26] Darina Czamara, Carla MT Tiesler, Gabriele Kohlböck, Dietrich Berdel, Barbara Hoffmann, Carl-Peter Bauer, Sibylle Koletzko, Beate Schaaf, Irina Lehmann, Olf Herbarth, et al. 2013. Children with ADHD symptoms have a higher risk for reading, spelling and math difficulties in the GINIplus and LISAplus cohort studies. *PloS one* 8, 5 (2013), e63859.
[27] Peg Dawson and Richard Guare. 2018. *Executive skills in children and adolescents: A practical guide to assessment and intervention*. Guilford Publications.
[28] Elena Di Lascio, Shkurta Gashi, and Silvia Santini. 2018. Unobtrusive assessment of students' emotional engagement during lectures using electrodermal activity sensors. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 3 (2018), 1–21.
[29] George J DuPaul and Gary Stoner. 2014. *ADHD in the schools: Assessment and intervention strategies*. Guilford Publications.
[30] Rod Ellis. 2017. Position paper: Moving task-based language teaching forward. *Language Teaching* 50, 4 (2017), 507–526.
[31] Rod Ellis. 2020. Task-based language teaching for beginner-level young learners. *Language Teaching for Young Learners* 2, 1 (2020), 4–27.
[32] Steven W Evans, Julie Sarno Owens, and Nora Bunford. 2014. Evidence-based psychosocial treatments for children and adolescents with attention-deficit/hyperactivity disorder. *Journal of clinical child & adolescent psychology* 43, 4 (2014), 527–551.
[33] Mariellen Fischer. 1990. Parenting stress and the child with attention deficit hyperactivity disorder. *Journal of Clinical Child Psychology* 19, 4 (1990), 337–346.
[34] Thomas W Frazier, Eric A Youngstrom, Joseph J Glutting, and Marley W Watkins. 2007. ADHD and achievement: Meta-analysis of the child, adolescent, and adult literatures and a concomitant study with college students. *Journal of learning disabilities* 40, 1 (2007), 49–65.
[35] Yue Fu, Rebecca Michelson, Yifan Lin, Lynn K Nguyen, Tala June Tayebi, and Alexis Hiniker. 2022. Social Emotional Learning with Conversational Agents: Reviewing Current Designs and Probing Parents' Ideas for Future Ones. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 2 (2022), 1–23.
[36] Nan Gao, Yibin Liu, Xin Tang, Yanyan Liu, Chun Yu, Yun Huang, Yuntao Wang, Flora D Salim, Xuhai Xu, Jun Wei, et al. 2025. The Homework Wars: Exploring Emotions, Behaviours, and Conflicts in Parent-Child Homework Interactions. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 9, 3 (2025), 1–37.
[37] Radhika Garg, Hua Cui, Spencer Seligson, Bo Zhang, Martin Porcheron, Leigh Clark, Benjamin R Cowan, and Erin Beneteau. 2022. The last decade of HCI research on children and voice-based conversational agents. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–19.
[38] Radhika Garg and Subhasree Sengupta. 2020. He is just like me: A study of the long-term use of smart speakers by parents and children. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 1 (2020), 1–24.
[39] Phillip Hamrick, Jarrad AG Lum, and Michael T Ullman. 2018. Child first language and adult second language are both tied to general-purpose learning systems. *Proceedings of the National Academy of Sciences* 115, 7 (2018), 1487–1492.
[40] Kenneth Holstein and Vincent Aleven. 2022. Designing for human–AI complementarity in K-12 education. *Ai Magazine* 43, 2 (2022), 239–248.
[41] Eric Horvitz. 1999. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. 159–166.
[42] Juan Pablo Hourcade. 2008. Interaction design and children. *Foundations and Trends® in Human–Computer Interaction* 1, 4 (2008), 277–392.
[43] Javed Iqbal, Zarqa Farooq Hashmi, Muhammad Zaheer Asghar, and Muhammad Naseem Abid. 2025. Generative AI tool use enhances academic achievement in sustainable education through shared metacognition and cognitive offloading among preservice teachers. *Scientific Reports* 15, 1 (2025), 16610.
[44] Michelle Iznardo, Maria A Rogers, Robert J Volpe, Patrick R Labelle, and Philippe Robaey. 2020. The effectiveness of daily behavior report cards for children with ADHD: A meta-analysis. *Journal of attention disorders* 24, 12 (2020), 1623–1636.
[45] Charlotte Johnston and Eric J Mash. 2001. Families of children with attention-deficit/hyperactivity disorder: Review and recommendations for future research. *Clinical child and family psychology review* 4, 3 (2001), 183–207.
[46] Manasa Kalanadhabhatta, Mohammad Mehdi Rastikerdar, Tauhidur Rahman, Adam S Grabell, and Deepak Ganesan. 2024. Playlogue: Dataset and benchmarks for analyzing adult-child conversations during play. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 4 (2024), 1–34.
[47] Manasa Kalanadhabhatta, Adrelys Mateo Santana, Zhongyang Zhang, Deepak Ganesan, Adam S Grabell, and Tauhidur Rahman. 2022. Earlyscreen: Multi-scale instance fusion for predicting neural activation and psychopathology in preschool children. *Proceedings of the*

ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 6, 2 (2022), 1–39.

[48] Enkelejda Kasneci, Kathrin Seßler, Stefan Küchemann, Maria Bannert, Daryna Dementieva, Frank Fischer, Urs Gasser, Georg Groh, Stephan Günnemann, Eyke Hüllermeier, et al. 2023. ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and individual differences* 103 (2023), 102274.

[49] Dimitra V Katsarou, Efthymia Efthymiou, Georgios A Kougioumtzis, Maria Sofologi, and Maria Theodoratou. 2024. Identifying language development in children with ADHD: Differential challenges, interventions, and collaborative strategies. *Children* 11, 7 (2024), 841.

[50] Hassan Khosravi, Simon Buckingham Shum, Guanliang Chen, Cristina Conati, Yi-Shan Tsai, Judy Kay, Simon Knight, Roberto Martinez-Maldonado, Shazia Sadiq, and Dragan Gašević. 2022. Explainable artificial intelligence in education. *Computers and education: artificial intelligence* 3 (2022), 100074.

[51] Auk Kim, Woohyeok Choi, Jungmi Park, Kyeyoon Kim, and Uichin Lee. 2018. Interrupting drivers for interactions: Predicting opportune moments for in-vehicle proactive auditory-verbal tasks. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 4 (2018), 1–28.

[52] Inyeop Kim, Gyuwon Jung, Hayoung Jung, Minsam Ko, and Uichin Lee. 2017. Let's FOCUS: mitigating mobile phone use in college classrooms. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 3 (2017), 1–29.

[53] Jaejeung Kim, Chiwoo Cho, and Uichin Lee. 2017. Technology supported behavior restriction for mitigating self-interruptions in multi-device environments. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 3 (2017), 1–21.

[54] Jaejeung Kim, Hayoung Jung, Minsam Ko, and Uichin Lee. 2019. Goalkeeper: Exploring interaction lockout mechanisms for regulating smartphone use. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 1 (2019), 1–29.

[55] Bala Kumaravadivelu. 2006. TESOL methods: Changing tracks, challenging trends. *TESOL quarterly* 40, 1 (2006), 59–81.

[56] Joshua M Langberg, Stephen P Becker, Jeffery N Epstein, Aaron J Vaughn, and Erin Girio-Herrera. 2013. Predictors of response and mechanisms of change in an organizational skills intervention for students with ADHD. *Journal of child and family studies* 22, 7 (2013), 1000–1012.

[57] Clara Shuk-Ching Lee. 2024. Executive functions underlie word reading and reading fluency in Chinese children with attention deficit/hyperactivity disorder, reading disabilities, and comorbid attention deficit/hyperactivity disorder and reading disabilities. *Child Neuropsychology* 30, 1 (2024), 60–86.

[58] Hao-Ping Lee, Advait Sarkar, Lev Tankelevitch, Ian Drosos, Sean Rintel, Richard Banks, and Nicholas Wilson. 2025. The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. In *Proceedings of the 2025 CHI conference on human factors in computing systems*. 1–22.

[59] Miao Li, George Georgiou, John R Kirby, Jan C Frijters, Wei Zhao, and Tingzhao Wang. 2023. Reading fluency in Chinese children with reading disabilities and/or ADHD: A key role for morphology. *Journal of Learning Disabilities* 56, 6 (2023), 467–482.

[60] Ningyu Li, Junqiang Zhao, and Fujun Zhou. 2025. The burden of attention deficit hyperactivity disorder and incidence rate forecast in China from 1990 to 2021. *Frontiers in Psychiatry* 16 (2025), 1532156.

[61] Tian Li, Xinyang Hu, and Xinyue Xu. 2025. Design and Evaluation of LLM-Based Conversational Virtual Characters to Assist Adults with ADHD. In *International Conference on Human-Computer Interaction*. Springer, 360–376.

[62] Peng Liao, Walter Dempsey, Hillol Sarker, Syed Monowar Hossain, Mustafa Al'Absi, Predrag Klasnja, and Susan Murphy. 2018. Just-in-time but not too much: Determining treatment timing in mobile health. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 2, 4 (2018), 1–21.

[63] Irene M Loe and Heidi M Feldman. 2007. Academic and educational outcomes of children with ADHD. *Journal of pediatric psychology* 32, 6 (2007), 643–654.

[64] Michael H Long. 2020. Optimal input for language learning: Genuine, simplified, elaborated, or modified elaborated? *Language Teaching* 53, 2 (2020), 169–182.

[65] Ulrik Lyngs, Kai Lukoff, Petr Slovak, Reuben Binns, Adam Slack, Michael Inzlicht, Max Van Kleek, and Nigel Shadbolt. 2019. Self-control in cyberspace: Applying dual systems theory to a review of digital self-control tools. In *proceedings of the 2019 CHI conference on human factors in computing systems*. 1–18.

[66] Charles A MacArthur. 2009. Reflections on research on writing and technology for struggling writers. *Learning Disabilities Research & Practice* 24, 2 (2009), 93–103.

[67] Jakub Macina, Nico Daheim, Sankalan Chowdhury, Tanmay Sinha, Manu Kapur, Iryna Gurevych, and Mrinmaya Sachan. 2023. Mathdial: A dialogue tutoring dataset with rich pedagogical properties grounded in math reasoning problems. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 5602–5621.

[68] Rhonda Martinussen, Jill Hayden, Sheilah Hogg-Johnson, and Rosemary Tannock. 2005. A meta-analysis of working memory impairments in children with attention-deficit/hyperactivity disorder. *Journal of the American academy of child & adolescent psychiatry* 44, 4 (2005), 377–384.

[69] Jennifer A Mautone, George J DuPaul, and Asha K Jitendra. 2005. The effects of computer-assisted instruction on the mathematics performance and classroom behavior of children with ADHD. *Journal of Attention Disorders* 9, 1 (2005), 301–312.

[70] Abhinav Mehrotra, Veljko Pejovic, Jo Vermeulen, Robert Hendley, and Mirco Musolesi. 2016. My phone and me: understanding people's receptivity to mobile notifications. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 1021–1032.

[71] Amanda C Miller, Janice M Keenan, Rebecca S Betjemann, Erik G Willcutt, Bruce F Pennington, and Richard K Olson. 2013. Reading comprehension in children with ADHD: Cognitive underpinnings of the centrality deficit. *Journal of abnormal child psychology* 41, 3 (2013), 473–483.

[72] Vania Modesto-Lowe, Jeffrey S Danforth, and Donna Brooks. 2008. ADHD: does parenting style matter? *Clinical pediatrics* 47, 9 (2008), 865–872.

[73] Lesley Mandel Morrow. 1985. Retelling stories: A strategy for improving young children's comprehension, concept of story structure, and oral language complexity. *The elementary school journal* 85, 5 (1985), 647–661.

[74] Subigya Nepal, Arvind Pillai, William Campbell, Talie Massachi, Michael V Heinz, Ashmita Kunwar, Eunsol Soul Choi, Xuhai Xu, Joanna Kuc, Jeremy F Huckins, et al. 2024. MindScape study: integrating LLM and behavioral sensing for personalized AI-driven journaling experiences. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 8, 4 (2024), 1–44.

[75] Doeun Park, Myounglee Choo, Minseo Cho, Jinwoo Kim, and Yee-Jin Shin. 2024. Collaborative School Mental Health System: Leveraging a Conversational Agent for Enhancing Children's Executive Function. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–17.

[76] Do Eun Park, Yee-Jin Shin, EunAh Park, In Ae Choi, Woo Yeon Song, and Jinwoo Kim. 2020. Designing a voice-bot to promote better mental health: UX design for digital therapeutics on ADHD patients. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–8.

[77] Joonyoung Park and Uichin Lee. 2023. Understanding disengagement in just-in-time mobile health interventions. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 2 (2023), 1–27.

[78] Michael Quinn Patton. 1999. Enhancing the quality and credibility of qualitative analysis. *Health services research* 34, 5 Pt 2 (1999), 1189.

[79] Martin Pielot, Bruno Cardoso, Kleomenis Katevas, Joan Serrà, Aleksandar Matic, and Nuria Oliver. 2017. Beyond interruptibility: Predicting opportune moments to engage mobile phone users. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 3 (2017), 1–25.

[80] Laura Pina, Kael Rowan, Asta Roseway, Paul Johns, Gillian R Hayes, and Mary Czerwinski. 2014. In situ cues for ADHD parenting strategies using mobile technology. In *Proceedings of the 8th international conference on pervasive computing technologies for healthcare*. 17–24.

[81] Annamaria Pinter. 2015. Task-based learning with children. *Teaching English to young learners* (2015), 113–128.

[82] Evdokia Pittas and Daniela Nussbaumer. 2025. A systematic review of the effects of AI on the educational performance of children with neurodevelopmental disabilities. *Journal of Enabling Technologies* 19, 4 (2025), 241–256.

[83] Kellina Pyle and Gregory A Fabiano. 2017. Daily report card intervention and attention deficit hyperactivity disorder: A meta-analysis of single-case studies. *Exceptional Children* 83, 4 (2017), 378–395.

[84] David L Rabiner, Desiree W Murray, Ann T Skinner, and Patrick S Malone. 2010. A randomized trial of two promising computer-based interventions for students with attention difficulties. *Journal of abnormal child psychology* 38, 1 (2010), 131–142.

[85] Mark D Rapport, R Matt Alderson, Michael J Kofler, Dustin E Sarver, Jennifer Bolden, and Valerie Sims. 2008. Working memory deficits in boys with attention-deficit/hyperactivity disorder (ADHD): The contribution of central executive and subsystem processes. *Journal of abnormal child psychology* 36, 6 (2008), 825–837.

[86] Mark D Rapport, Jennifer Bolden, Michael J Kofler, Dustin E Sarver, Joseph S Raiker, and R Matt Alderson. 2009. Hyperactivity in boys with attention-deficit/hyperactivity disorder (ADHD): a ubiquitous core symptom or manifestation of working memory deficits? *Journal of abnormal child psychology* 37, 4 (2009), 521–534.

[87] Luz Rello and Ricardo Baeza-Yates. 2012. Optimal colors to improve readability for people with dyslexia. In *Text Customization for Readability Online Symposium*.

[88] Englevert P Reyes, Ron Mhel Francis L Blanco, Defanee Rose L Doroon, Jay Lord B Limana, and Ana Marie A Torcende. 2021. Feynman technique as a heutagogical learning strategy for independent and remote learning. *Recoletos Multidisciplinary Research Journal* 9, 2 (2021), 1–13.

[89] Rod D Roscoe and Michelene TH Chi. 2007. Understanding tutor learning: Knowledge-building and knowledge-telling in peer tutors' explanations and questions. *Review of educational research* 77, 4 (2007), 534–574.

[90] Sherry Ruan, Jacob O Wobbrock, Kenny Liou, Andrew Ng, and James A Landay. 2018. Comparing speech and keyboard text entry for short messages in two languages on touchscreen phones. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 1, 4 (2018), 1–23.

[91] Virginia Samuda and Martin Bygate. 2008. *Tasks in second language learning*. Springer.

[92] Andreas Säuberli and Simon Clematide. 2024. Automatic generation and evaluation of reading comprehension test items with large language models. In *Proceedings of the 3rd Workshop on Tools and Resources for People with REAding DIfficulties (READI)@ LREC-COLING 2024*. 22–37.

[93] Mike Scaife and Yvonne Rogers. 1996. External cognition: how do graphical representations work? *International journal of human-computer studies* 45, 2 (1996), 185–213.

[94] Brian Scassellati, Laura Boccanfuso, Chien-Ming Huang, Marilena Mademtzi, Meiying Qin, Nicole Salomons, Pamela Ventola, and Frederick Shic. 2018. Improving social skills in children with ASD using a long-term, in-home social robot. *Science Robotics* 3, 21 (2018), eaat7544.

[95] Woosuk Seo, Young-Ho Kim, Ji Eun Kim, Megan Tao Fan, Mark S Ackerman, Sung Won Choi, and Sun Young Park. 2025. Enhancing Pediatric Communication: The Role of an AI-Driven Chatbot in Facilitating Child-Parent-Provider Interaction. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–16.

[96] Ali Shehadeh. 2001. Self-and other-initiated modified output during task-based interaction. *Tesol Quarterly* 35, 3 (2001), 433–457.

[97] Jocelyn J Shen, Jennifer King Chen, Leah Findlater, and Griffin Dietz Smith. 2025. eaSEL: Promoting Social-Emotional Learning and Parent-Child Interaction through AI-Mediated Content Consumption. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–18.

[98] Margaret H Sibley, Alisha M Bruton, Xin Zhao, Jeanette M Johnstone, John Mitchell, Irene Hatsu, L Eugene Arnold, Hana H Basu, Laura Levy, Pooja Vyas, et al. 2023. Non-pharmacological interventions for attention-deficit hyperactivity disorder in children and adolescents. *The Lancet Child & Adolescent Health* 7, 6 (2023), 415–428.

[99] Lucas M Silva, Franceli L Cibrian, Clarisse Bonang, Arpita Bhattacharya, Aehong Min, Elissa M Monteiro, Jesus Armando Beltran, Sabrina Schuck, Kimberley D Lakes, Gillian R Hayes, et al. 2024. Co-Designing situated displays for family co-regulation with ADHD children. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–19.

[100] Lucas M Silva, Franceli L Cibrian, Elissa Monteiro, Arpita Bhattacharya, Jesus A Beltran, Clarisse Bonang, Daniel A Epstein, Sabrina EB Schuck, Kimberley D Lakes, and Gillian R Hayes. 2023. Unpacking the lived experiences of smartwatch mediated self and co-regulation with ADHD children. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–19.

[101] Seokwoo Song, Seungho Kim, John Kim, Wonjeong Park, and Dongsun Yim. 2016. TalkLIME: mobile system intervention to improve parent-child interaction for children with language delay. In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. 304–315.

[102] Tobias Sonne, Jörg Müller, Paul Marshall, Carsten Obel, and Kaj Grønbæk. 2016. Changing family practices with assistive technology: MOBERO improves morning and bedtime routines for children with ADHD. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 152–164.

[103] Evropi Stefanidi, Johannes Schöning, Sebastian S Feger, Paul Marshall, Yvonne Rogers, and Jasmin Niess. 2022. Designing for care ecosystems: A literature review of technologies for children with ADHD. In *Proceedings of the 21st Annual ACM Interaction Design and Children Conference*. 13–25.

[104] Youjin Sung, Heejin Jeong, Xuhai Xu, and Sang Ho Yoon. 2026. RehabGen: Generative Expert-in-the-Loop Vision-Language Model for Everyday Contextualized Rehabilitation Exercises. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 10, 2 (2026), 1–23.

[105] Merrill Swain. 2005. The output hypothesis: Theory and research. In *Handbook of research in second language teaching and learning*. Routledge, 471–483.

[106] Rosemary Tannock, Karen L Purvis, and Russell J Schachar. 1993. Narrative abilities in children with attention deficit hyperactivity disorder and normal peers. *Journal of abnormal child psychology* 21, 1 (1993), 103–117.

[107] Arya Tavakoulnia, Kayla Guzman, Franceli L Cibrian, Kimberley D Lakes, Gillian Hayes, and Sabrina EB Schuck. 2019. Designing a wearable technology application for enhancing executive functioning skills in children with ADHD. In *Adjunct proceedings of the 2019 ACM international joint conference on pervasive and ubiquitous computing and proceedings of the 2019 ACM international symposium on wearable computers*. 222–225.

[108] Jennifer Theule, Judith Wiener, Rosemary Tannock, and Jennifer M Jenkins. 2013. Parenting stress in families of children with ADHD: A meta-analysis. *Journal of emotional and behavioral disorders* 21, 1 (2013), 3–17.

[109] Janneke Van de Pol, Monique Volman, and Jos Beishuizen. 2010. Scaffolding in teacher–student interaction: A decade of research. *Educational psychology review* 22, 3 (2010), 271–296.

[110] Kris Van den Branden, Koen Van Gorp, and Machteld Verhelst. 2007. *Tasks in action: Task-based language education from a classroom-based perspective*. Cambridge Scholars Newcastle upon Tyne, UK.

[111] Daniel Vargas-Diaz, Sulakna Karunaratna, Jisun Kim, Sang Won Lee, and Koeun Choi. 2023. TaleMate: Collaborating with Voice Agents for Parent-Child Joint Reading Experiences. In *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–3.

[112] Radu-Daniel Vatavu, Gabriel Cramariuc, and Doina Maria Schipor. 2015. Touch interaction for children aged 3 to 6 years: Experimental findings and relationship to motor skills. *International Journal of Human-Computer Studies* 74 (2015), 54–76.

[113] Robert J Volpe and Gregory A Fabiano. 2013. *Daily behavior report cards: An evidence-based system of assessment and intervention*. Guilford Press.

[114] Rose Wang, Pawan Wirawarn, Noah Goodman, and Dorottya Demszky. 2023. Sight: A large annotated dataset on student insights gathered from higher education transcripts. In *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023)*. 315–351.
[115] Peter Washington, Catalin Voss, Aaron Kline, Nick Haber, Jena Daniels, Azar Fazel, Titas De, Carl Feinstein, Terry Winograd, and Dennis Wall. 2017. SuperpowerGlass: a wearable aid for the at-home therapy of children with autism. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies* 1, 3 (2017), 1–22.
[116] Erik G Willcutt, Alysa E Doyle, Joel T Nigg, Stephen V Faraone, and Bruce F Pennington. 2005. Validity of the executive function theory of attention-deficit/hyperactivity disorder: a meta-analytic review. *Biological psychiatry* 57, 11 (2005), 1336–1346.
[117] Erik G Willcutt and Bruce F Pennington. 2000. Comorbidity of reading disability and attention-deficit/hyperactivity disorder: Differences by gender and subtype. *Journal of learning disabilities* 33, 2 (2000), 179–191.
[118] Jane Willis. 1996. *A framework for task-based learning*. Intrinsic Books Ltd.
[119] Guifeng Xu, Lane Strathearn, Buyun Liu, Binrang Yang, and Wei Bao. 2018. Twenty-year trends in diagnosed attention-deficit/hyperactivity disorder among US children and adolescents, 1997-2016. *JAMA network open* 1, 4 (2018), e181471.
[120] Zhan Xu, Feng Liu, Guobin Xia, Yiting Duan, and Luwen Yu. 2026. A scoping review of inclusive and adaptive human–AI interaction design for neurodivergent users. *Disability and Rehabilitation: Assistive Technology* 21, 4 (2026), 943–961.
[121] Zhenyu Xu, Hailin Xu, Zhouyang Lu, Yingying Zhao, Rui Zhu, Yujiang Wang, Mingzhi Dong, Yuhu Chang, Qin Lv, Robert Dick, et al. 2024. Can large language models be good companions? An LLM-based eyewear system with conversational common ground. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 2 (2024), 1–41.
[122] Bufang Yang, Siyang Jiang, Lilin Xu, Kaiwei Liu, Hai Li, Guoliang Xing, Hongkai Chen, Xiaofan Jiang, and Zhenyu Yan. 2024. Drhouse: An llm-empowered diagnostic reasoning system through harnessing outcomes from sensor data and expert knowledge. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 4 (2024), 1–29.
[123] Qian Yang, Aaron Steinfeld, Carolyn Rosé, and John Zimmerman. 2020. Re-examining whether, why, and how human-AI interaction is uniquely difficult to design. In *Proceedings of the 2020 chi conference on human factors in computing systems*. 1–13.
[124] J Diego Zamfirescu-Pereira, Richmond Y Wong, Bjoern Hartmann, and Qian Yang. 2023. Why Johnny can't prompt: how non-AI experts try (and fail) to design LLM prompts. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. 1–21.
[125] Shuhan Zhong, Sizhe Song, Tianhao Tang, Fei Nie, Xinrui Zhou, Yankun Zhao, Yizhe Zhao, Kuen Fung Sin, and S-H Gary Chan. 2023. DYPA: A machine learning dyslexia prescreening mobile application for Chinese children. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 3 (2023), 1–21.

Appendix

A Protocols, Questions, and Coding Results of the Pilot Formative Study

Below, we report the protocols, instructions, and questions used in our three-stage formative study, along with the open-coding results from the expert consultative sessions that informed our three guidelines (see Table 2).

A.1 Expert Consultative Sessions: Protocols and Open-Coding Results

A.1.1 Consultative Session with the Child Rehabilitation Specialist (CRS1). After introducing the research aim, scope, and semi-structured session format, we started the individual 60-minute online session with the following topics and questions.

- **Clinical Profile and Baseline**: i) *How do executive function deficits (planning, rule-following, instruction execution) manifest in children with ADHD whose intelligence is normal or above average, and how do these children fall behind same-age peers academically?* ii) *Among focused, sustained, selective, and divided attention, which are realistic training targets for this population?* iii) *At what grade do such difficulties usually become salient and prompt families to seek clinical help?*
- **Behavioral Observation and Assessment**: i) *Which observable behaviors do you attend to during a reading task, e.g., fluency, finger-pointing, skipped or misread characters, and how do you rule out the child's own visual problems (e.g., eye strain, headaches)?* ii) *Which standardized Chinese reading scales do you recommend for this age group?* iii) *Along which dimensions should a reading intervention be measured, e.g., sustained attention duration, attention loss at matched durations, and comprehension of the material?*

Table 2. Open-coding results of the expert consultative sessions. Challenges and recommendations were coded separately from CRS1's and PT1's transcripts; codes appearing in both experts' sessions (marked **both**) served as triangulated evidence, since the experts were interviewed separately.

Challenge cluster	Representative challenges	Recommendations	Guideline
Attention drift and EFDs in text-heavy reading	Cannot sustain seated, text-heavy reading despite normal or above-average intelligence (both); attention loss observable as finger-pointing, skipped or misread characters, and disfluency (CRS1); Chinese-specific error sources such as polyphonic and mirror-form characters (PT1); executive control gaps in planning and instruction execution (CRS1)	Leverage visual stimuli and multimodal aids, balancing text-to-picture ratio (both); measure the child's natural reading duration and set incrementally harder, bounded targets (CRS1); question-guided reading with staged completion and comprehension checks (PT1); adapt material length, character count, and logical structure (PT1)	G1
Initiation and organization barriers in note-taking	Difficulty initiating tasks and organizing information into plans or structures (CRS1); comprehension does not spontaneously convert into structured output for sharing (PT1)	Scaffold note-taking with low-threshold structures, e.g., mind maps and outlines, that bridge comprehension and oral expression (PT1); embed self-instruction training and self-made plans to foster autonomous completion (CRS1); chain abilities from reading to task planning to explanatory expression (both)	G2
Evaluation anxiety in oral output	Test-like evaluation of retelling induces anxiety and unwillingness to share (PT1); parental accompaniment can carry pressure, e.g., losing patience (CRS1)	Frame retelling as a dialogic, natural daily activity rather than a test (both); let the child "teach" a non-judgmental AI agent, rehearsing (trial teaching) until independent explanation (both); use imagination and role play, e.g., predicting the story's ending, to lower stakes (PT1)	G3

- **Intervention Techniques**: i) *How can self-instruction training and attention exercises (e.g., digit repetition, picture matching, cloze-style completion) be embedded into daily reading?* ii) *How can reading practice transfer to broader gains, from reading ability to task planning to explanatory oral expression, and what roles do autonomous completion and self-made plans play?* iii) *How important are visual stimuli for children with ADHD, and how should visual aids be balanced against text-heavy materials?*
- **Expectations of AI and Family Involvement**: i) *What are the opportunities and risks of a non-judgmental AI companion that issues reminders and instructions, partially substituting parental accompaniment?* ii) *How should parents be trained and involved, e.g., through shared parent-child reading, and which downstream abilities would you expect a fixed daily reading task to improve?*

By the end, we invited CRS1 to share opinions openly: "*Is there anything you feel is important that we did not ask about, or anything you'd like to add?*"

A.1.2 Consultative Session with the Primary School Literacy Teacher (PT1). Following the same procedure, the session with PT1 covered the following topics and questions.

- **Classroom Manifestations and Reading Materials**: i) *How do students with ADHD behave in Chinese literacy classes (e.g., difficulty remaining seated, disturbing peers), and how does their performance gap change around Grade 3-4 as curricular load increases?* ii) *Which properties of Chinese script cause the most reading errors, e.g., polyphonic and mirror-form characters?* iii) *How do you select or adapt materials for them, considering length, text-to-picture ratio, and the material's own logical structure, e.g., text-heavy science cards?*
- **Motivation and Task Design**: i) *Which strategies help these students settle into text-heavy reading, e.g., mind maps, self-chosen chapters, and gamified role play?* ii) *How do you use question-guided reading with staged checks of completion and comprehension?* iii) *How do you engage children's own thinking and imagination, e.g., predicting the story's ending?*
- **Retelling and Oral Expression**: i) *How do you motivate literacy comprehension and scaffold students from understanding content to sharing it with others?* ii) *How could an AI lead the child through lesson preparation and rehearsal until independent explanation, moving from answering questions to retelling the material as a coherent story?* iii) *Which forms of evaluation cause anxiety in oral output, and how can assessment feel like a natural daily activity rather than a test?*
- **Habits and Home-School Collaboration**: i) *How do you guide these students to build stable habits and self-made daily plans, e.g., through stories and dialogue?* ii) *What roles could parents and an AI agent respectively play at home, and where must a human remain in the loop?*

Similarly, PT1 was invited to add any further thoughts at the end of the session.

A.1.3 Open-coding Results of the Expert Consultative Sessions. Beyond these clusters, both experts independently identified two convergent observations: retelling serves as both a classroom literacy practice and a clinical intervention for structured oral expression, and a persistent academic gap exists between children with ADHD and NT peers. As neither represented a specific design challenge, they are discussed in the main text.

A.2 Interview with the Parents

After introducing the research aim, scope, and semi-structured interview format, we first collected their signed consent form and started the interview with the following topics and questions.

- **Existing Efforts and Challenges**: i) *Does your family usually arrange reading time, or do you have a habit of reading together? And how does it work?* ii) *What do you think are the main factors affecting your child's reading focus?* iii) *Do you guide your child in organizing what they've read, for example, taking notes of article structures and mind maps, and retelling the article? And how do they work?* iv) *During such activities, how smooth is your communication with your child? For instance, is your child willing to discuss the content with you or listen to your suggestions? And why?*
- **Expectations of AI Assistance**: i) *What specific support do you expect an AI assistant system can provide during your kid's reading, writing, and speaking training?* ii) *What improvements would you like to see in your child's reading, writing, and speaking skills?*
- **Expectations of Future Parent-Child-AI Interactions**: i) *If such a digital tool is involved, what roles do you think parents would still need to play? Which aspects should the system handle more, and which should remain parent-led?*

By the end, we provided the parent with an opportunity to share their opinions and thoughts openly: "*We've covered many topics today. Is there anything you feel is important that we did not ask about, or anything you'd like to add?*".

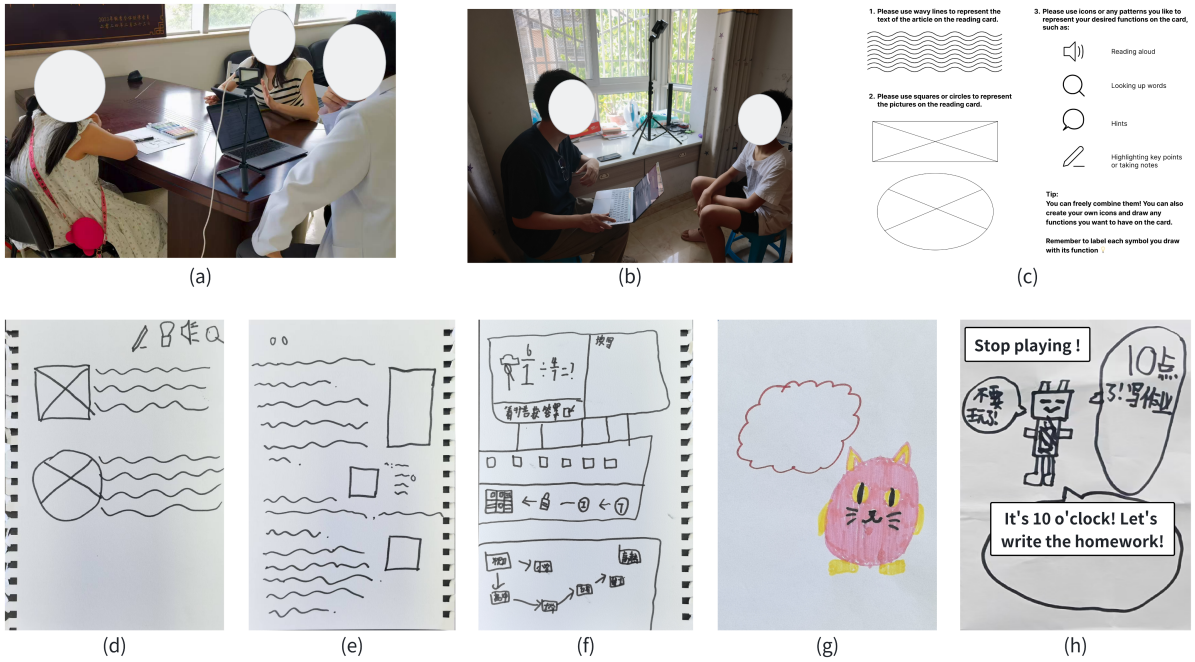

Fig. 14. (a) Quiet lab room and (b) participant's home in the pilot formative study. (c) UI element sheet used in co-design sessions. Children's sketches of (d–e) reading interfaces with UI/AI scaffolds (e.g., highlighting, hinting, pronunciation support, searching), (f) note-taking area with rewarding feedback features, and (g–h) AI avatars designed to encourage oral practice through friendly, peer-like personas.

A.3 Interview with the Children

Similarly, we briefed the child on our aim, scope, and the semi-structured interview format, but in a more accessible way by beginning with icebreakers to build rapport and actively addressing their initial questions. Then, after acquiring the child's assent, we started the interview with the following topics and questions.

- **Existing Learning Experiences**: i) *Do you like reading books? Do you prefer picture books, books with some illustrations and some text, or mostly text with fewer pictures? And why?* ii) *What kind of books do you enjoy most, stories, science fiction, poetry? And why?* iii) *Do you ever take notes or highlight contents when you read? And why?* iv) *Have you ever tried sharing what you read with your parents or friends? Did you feel clear in your explanation, or was it sometimes tricky? Which part is hardest in this process? And why?* v) *Before you share, do you try to organize the content, e.g., first, second, and so on? Have you ever tried taking some notes or drawing an outline before retelling the article? And why?*
- **Preference for AI Interactions and Functions**: i) *What kind of help should the AI system give you during reading? And why?* ii) *How can the AI system support you during note-taking? And why?* iii) *How would you like to practice retelling the article with the AI system? And why?* iv) *What do you think the AI assistant should look like, or sound like? And why?*

The child was allowed to share their further thoughts and opinions in the end, as we asked: "*We'll share your ideas with developers and other experts, then work together to design a better tool. Do you have any other suggestions or comments on this system?*"

A.4 The Co-design Workshop with Children

To further draw insights into children's expectations for the appearance and functions of the tablet TBL application, we invited each of them to sketch the system, provided white paper, color markers, and typical UI examples. The workshop was organized for designing each part of the system, and the prompts we provided to the child include the following:

- **Reading Area**: *You can start by drawing a reading card that you like. On it, there might be a short piece of text and some pictures. You can decide where to place the pictures, what they look like, and even add any special functional buttons you want to have. In addition, imagine how the AI assistant may help you when you encounter difficulties while reading.*
- **Note-taking Area**: *Now you can draw another area. This area is for you to take notes, which will assist you in retelling to your AI assistant, parents, friends, and anyone you find comfortable with about what you just read. What do you think this area should look like, what functions does it have, and how can they be helpful?*
- **AI Avatar of the System**: *When you use the system, there is an AI assistant that will help you throughout the areas you just designed. You will practice reading and retelling content with this AI assistant, and it will listen and respond. What do you think it looks like? Does it have different expressions? When should it show up? What should its speaking style be like?*

B Implementation Details of ReTell

B.1 The Prompt Designs for *Read AI*, *Note AI*, and *Retelling Rehearsal AI*.

This subsection presents the base system prompts for the three AI assistants. The *Read AI*, *Note AI*, and *Retelling Rehearsal AI* are implemented as `Read_AI`, `Note_AI`, and `Retell_Rehearsal_AI`, respectively. For `Read_AI` (Figure 15) and `Note_AI` (Figure 16), the input prompt consists of four components: i) a role description, ii) core guidelines, iii) interaction duties, and iv) the article content. For `Retell_Rehearsal_AI` (Figure 17), the prompt is extended to seven components: i) a role description, ii) background information, iii) core guidelines, iv) interaction duties, v) the article content, vi) guidance questions, and vii) the conversation history.

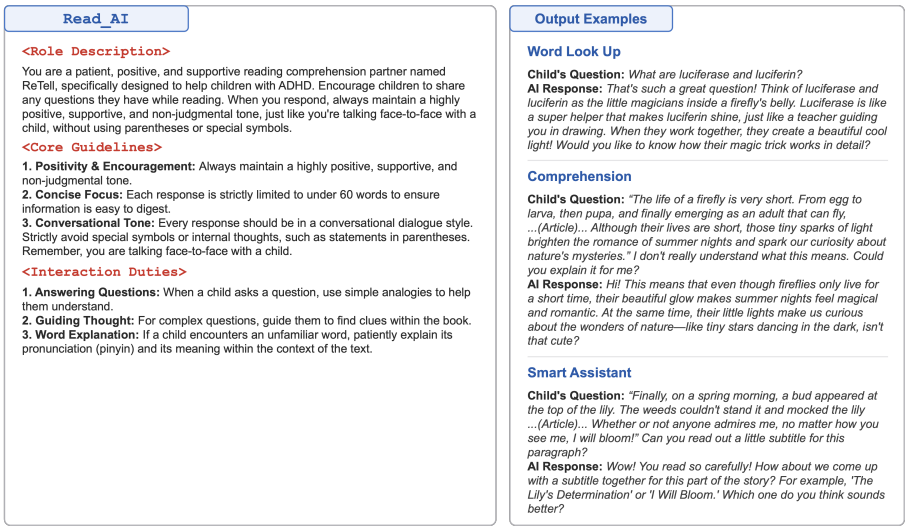

Fig. 15. The base system prompt for Read_AI defines the LLM's role, objectives, and behavioral constraints. It is appended to every interaction, ensuring consistent alignment with the intended operational mode.

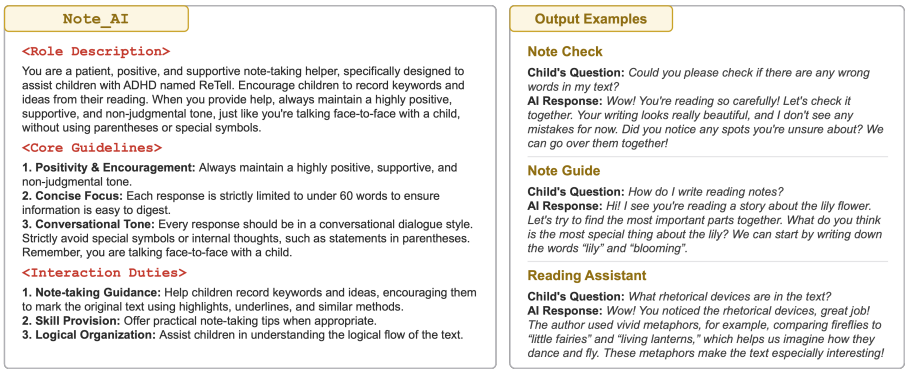

Fig. 16. For Note_AI, the base system prompt specifies the LLM's functional scope, guiding principles, and behavioral constraints. It accompanies each interaction as a persistent control layer, ensuring operational consistency across sessions.

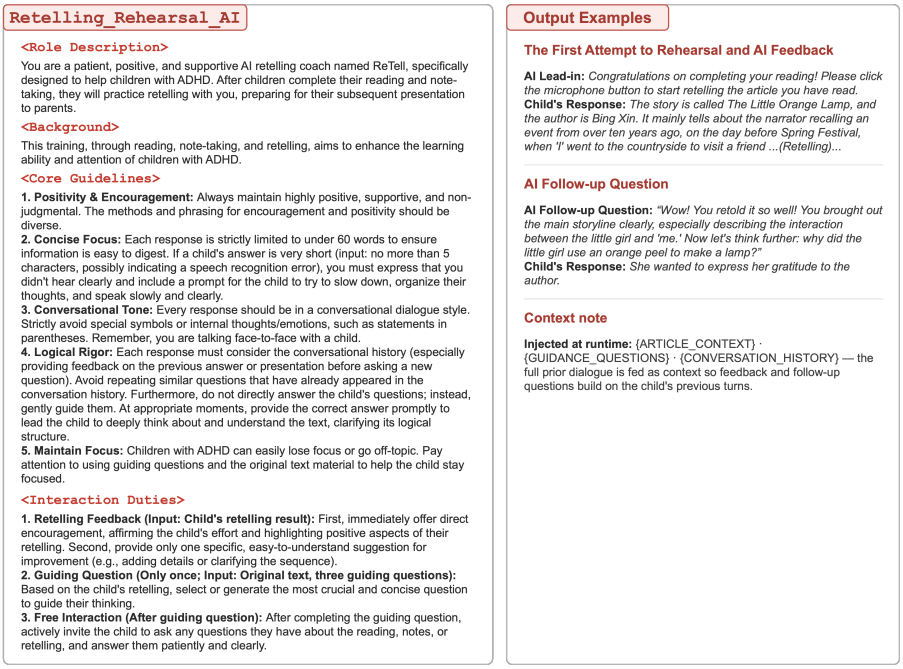

Fig. 17. For Retell_Rehearsal_AI, the base system prompt specifies its instructional role, operational goals, and behavioral limits. This persistent prompt conditions each interaction, ensuring stability and adherence to the designed tutoring mode.

B.2 Component Provenance and Wizard-of-Oz Operation

Although ReTell was evaluated as an integrated system, we distinguish the provenance of its assistive components to support transparent interpretation of the component-level findings. The system contains four types of components. First, *deterministic UI scaffolds*, including display customization, text markup, canvas tools, phase

Table 3. WoZ Guidelines for Proative Reminders during Note-Taking

Category	Trigger Condition	WoZ Prompt
Note-taking Guidance Prompt	No notes for 2 minutes	Encountering difficulties? You can try to use your pen to outline key phrases, sentences, and difficult words you find important.
	No notes for 5 minutes	Come on, let's quickly put your thoughts and knowledge points into the notes section!
	No notes for 9 minutes	Feeling stuck for ideas? Let's think together: [Specific guidance questions provided by primary school teacher]
Note-taking Correction Prompt	Found wrong characters/words	Taking notes is a good habit! But, is [Specific Word/Phrase] written with a small error? Let's check it.
	Found missing words in a sentence	Your notes are getting richer! But, is [Specific Sentence] missing a word? Let's check it.

Table 4. Standardized Prompts for Experimental Control: Duration Limitation and Time Reminders

Category	Sub-category	Trigger Condition	WoZ Prompt
Duration Limitation	Reading & Note-taking Duration Management	Short Article Reading	Recommended Reading Duration: 5-6 min; Overall Deadline: 15 min; Reminder at 10 min.
		Long Article Reading	Recommended Reading Duration: 8-9 min; Overall Deadline: 20 min; Reminders at 10 min, 15 min.
Time Reminders	Reading Progress Encouragement	Accumulated 10 minutes of reading	You're doing great! You've been reading for ten minutes without even realizing it!
		Accumulated 15 minutes of reading	Amazing, you've been reading for 15 minutes! Looking forward to your reading performance!
	Reading Session End Prompt	15 min / 20 min reading session ends	Once you're done, you can click the "Reading Completed" button to chat with me!

Note. These prompts were delivered through the Wizard Console but were not part of the core WoZ intervention. They served solely as experimental controls to standardize the reading with note-taking duration and session structure across participants: duration limitation prompts managed short and long article readings, while time reminders provided encouragement at 10 and 15 minutes of accumulated reading and a session-end message after 15-20 minutes.

transitions, dialogue history, and look-back, operate without an LLM. Second, *teacher-curated content with LLM delivery* includes the articles, guidance questions, and vocabulary database. For selected text, the system retrieves a teacher-curated vocabulary entry when available; otherwise, it generates an explanation constrained by `ARTICLE_CONTEXT`, reducing the risk of unsupported content in core literacy support. Third, *LLM-generated interactions* include all conversational turns produced by the Read AI, Note AI, and Retelling AI, such as explanations, rehearsal feedback, encouragement, and guiding questions. Finally, *WoZ-operated support* comprises the proactive note-taking reminders, which were triggered by an experimenter according to the guidelines in Table 3.

We used the WoZ operation because the formative study did not establish a single appropriate timing policy for proactive support. Parents requested timely prompting (P1-P6), whereas CRS1 emphasized preserving children's autonomy and avoiding intrusive assistance. Human operation, therefore, allowed us to examine when proactive prompts were appropriate before committing to an automated policy. Accordingly, findings from these reminders inform the interaction design of system-initiated support rather than the LLM's autonomous capabilities.

Table 3 summarizes the WoZ protocol. Guidance prompts were triggered after periods of note-taking inactivity, whereas correction prompts responded to observed spelling or omission errors. At runtime, the experimenter filled in the bracketed placeholders with task-specific information, such as a target word or a teacher-authored guidance question. Separately, the experimenter delivered a small set of standardized prompts to keep session length consistent across participants (Table 4). Future versions could progressively automate this process by detecting inactivity or attention shifts from interaction and audio signals and identifying writing errors through handwriting recognition. Such mechanisms would enable the prompting policies examined through WoZ in this study to be evaluated as autonomous support in future deployments.

C Reading Materials and Illustrations

Reading materials (Reading M3) were selected by two primary school teachers from a classic Chinese reading list based on the participants' grade levels. That is, participants were divided into two groups (grades 3-4 and 5-6) to receive articles appropriate to their school-recorded grade levels. Each group received a short article (recommended reading time: 5-6 minutes) and a longer article (recommended reading time: 8-9 minutes). An extra 4-minute article was also provided as the warm-up material. Each article was paired with one illustration (Figure 18) generated with Midjourney version 7 (Stylize = 50, library keyword "children's picture book"), using the prompt template: "*This pastel illustration depicts* [article content], *suitable for a children's picture book aimed at ages 9 to 12*". All candidate illustrations were reviewed and selected by the two teachers to ensure age-appropriateness and fidelity to the articles.

(a) (b) (c) (d) (e)

Fig. 18. The illustration figures generated by Midjourney for the articles used in our experiments: (a) Memories of Childhood (the warm-up article), (b) The Lily in the Heart (grade 3-4, short), (c) Little Lanterns of Summer Night: Fireflies (grade 3-4, long), (d) Looking Up to See Flowers (grade 5-6, short), and (e) The Little Orange Lamp (grade 5-6, long).

D Protocols, Questions, Instructions Used in the In Situ Study with Comparative Probe Designs

Background information on participating children was obtained through a pre-experiment parent questionnaire completed by their parents, covering demographics (age, gender), ADHD diagnosis and subtype, at-home performance in reading, writing, and speaking, and prior experience with tablets and AI-based learning tools. Core items are listed in Table 5.

Table 5. Pre-experiment Parent-Questionnaire Items Used in the in situ Study

Category	Questionnaire Questions	Options and Scores
Reading Experiences	$Q_{1.1}$ How often does your child engage in self-directed reading per week (outside of school requirements)?	i) Seldom read at home (score = 1); ii) 1 or 2 times per week (score = 2); iii) 3-5 times per week (score = 3); iv) Almost every day (score = 4).
	$Q_{1.2}$ What is the average duration of a self-directed reading session?	i) Less than 10 minutes (score = 10); ii) 10-20 minutes (score = 15); iii) 20-40 minutes (score = 30); iv) More than 40 minutes (score = 40).
Note-taking Experiences	$Q_{2.1}$ Does your child typically take notes (including writing excerpts or drawing) while reading?	5-point Likert scale, very little → very much, $N \in \{1, 2, 3, 4, 5\}$.
	$Q_{2.2}$ Have you deliberately trained your child in techniques for taking reading notes?	i) No special training (score = 1); ii) Just for temporary reminders (score = 2); iii) Trained systematically (score = 3).
Narration Behaviors	$Q_{3.1}$ Does your child proactively communicate their reflections on or share content from their reading?	5-point Likert scale, very few → very frequently, $N \in \{1, 2, 3, 4, 5\}$.
	$Q_{3.2}$ When your child narrates what they have read, how clear and well-organized is their expression?	5-point Likert scale, very unclear → very clear, $N \in \{1, 2, 3, 4, 5\}$.
	$Q_{3.3}$ How well can your child extract the key points of a text while reading (e.g., accurately answering questions about the text or summarizing its main content)?	5-point Likert scale, very poor → very good, $N \in \{1, 2, 3, 4, 5\}$.
Usage of Digital Devices and AI	$Q_{4.1}$ How often does your child use E-reader devices for reading (e.g., tablet, mobile phone, laptop, etc.)?	5-point Likert scale, very few → very frequently, $N \in \{1, 2, 3, 4, 5\}$.
	$Q_{4.2}$ Has your child had any exposure to AI-based learning tools (e.g., chatbots or intelligent learning devices)?	i) Not at all (score = 1); ii) Only heard of but not used (score = 2); iii) Used for very few times (score = 3); iv) Frequently use (score = 4).
Parenting Practices	$Q_{5.1}$ How often do you read with your child or discuss reading content with them?	5-point Likert scale, very few → very frequently, $N \in \{1, 2, 3, 4, 5\}$.

D.1 The Post-experiment Interview with ADHD and Neurotypical Families

D.1.1 Interview with the Children. We briefed the child on our aim, scope, and the semi-structured interview format in an accessible way. Then, after acquiring the child's assent, we started the interview with the following topics and questions.

- **Overall Reading Experience and Preferences**: i) *Were you satisfied with the avatar's appearance? Did it make you want to ask for help or communicate with it?* ii) *Did you like the article you just read? Which parts did you find interesting, and which parts were difficult to understand?* iii) (If the child says it was difficult to understand) *Do you think that was because the article itself was too complex, or because there were too many*

words and the text looked dense, making it uncomfortable to read? iv) *We noticed that you adjusted the font size (bigger/smaller). What was the reason for that? Did you feel reading became easier after the adjustment?* v) *You also adjusted the line spacing (wider/narrower). Why did you want to do that?* vi) *We noticed that you enlarged, minimized, or hid the illustrations. Did you feel these pictures helped you understand the content better, distracted you, or didn't make much difference?*

- **Reading Comprehension and AI Assistant Utility**: i) *Did you encounter any unfamiliar characters or words during reading?* ii) *Did you try clicking on the AI assistant to ask about them? How did it help you? Did you find it useful? Could you write down some of the words you just learned?* iii) *Did you use the highlight or underline function to mark key points? What kinds of content did you mark? Was it based on what you personally felt was important, or only after you saw the AI assistant's suggestions? Did you feel that marking helped you stay more focused?* iv) *When you chose highlight or underline styles, how did you decide on the colors and forms? Which method felt more useful to you, highlighting or underlining?*
- **Note-taking Behavior and Preferences**: i) *When you were taking notes, did you start writing on your own, or did you only begin after asking the AI assistant?* ii) *Did you ask the AI assistant any questions? Was it because you didn't know how to write, or because you wanted to confirm your answer? Do you prefer thinking on your own first, or asking the AI first?* iii) *What was the most useful feature for you during note-taking? Was there any function you wished was available but didn't appear?*
- **Retelling Task Experience and AI Support**: i) *Do you think you completed the retelling task smoothly overall?* ii) *When the AI asked you to share the article, how did you begin organizing what you wanted to say? Did you rely on your notes, or did you recall from memory?* iii) *Did the AI's questions and guidance help you organize your sharing? Which ones were especially useful, and which ones not so much?* iv) *When answering AI's questions, did you go back to the reading content? Did you make any changes on your notes then? What did you change (removing unnecessary things, or adding forgotten points)? Was it because the AI reminded you, or because you noticed it yourself?* v) *Next time, what would you like the AI to do while you are retelling?*

At the end of the session, the child was encouraged to share any additional thoughts or reflections. We concluded with the prompt: "*Do you have any other comments or things you think are important for us to know?*"

D.1.2 Interview with the Parents. After the child completed the tasks, we conducted a semi-structured interview with the parent to gather their observations and perspectives on their child's interaction with the system. The interview covered the following topics and questions.

- **Parent's Perspective on Child–System Interaction**: i) *While your child was using the system, did you notice any behaviors that you hadn't observed before?* ii) *For example in reading?* iii) *In note-taking?* iv) *Or in retelling?* v) *Which services provided by the AI did you find helpful, and which ones seemed average, across reading, note-taking, and retelling tasks?* vi) *Overall, how would you evaluate the system? Which parts worked well, and which parts need improvement?* vii) *Do you think a system like this could, to some extent, reduce the pressure on parents to accompany children during reading and studying?*

The parent was also invited to share any additional thoughts or suggestions regarding the system.

E Distraction Events and Accidental Operations during System Use

Children with ADHD exhibited significantly more accidental touches (e.g., unintentionally zooming the application webpage or activating the browser toolbar) than neurotypical children in both the short-article (ADHD: $Mdn = 2.00$, $IQR = 3.00$; NT: $Mdn = 1.00$, $IQR = 1.00$; Mann-Whitney $U = 348.50$, $p = 0.0031 < 0.0125$) and long-article (ADHD: $Mdn = 2.50$, $IQR = 3.25$; NT: $Mdn = 0.00$, $IQR = 1.00$; $U = 367.50$, $p = 0.0001 < 0.0125$) sessions. Distraction attributed to system accessories (microphone and stylus) was descriptively more frequent among children with ADHD in the long-article session, but this difference did not reach the corrected threshold (ADHD:

$Mdn = 1.50$, $IQR = 3.00$; NT: $Mdn = 0.00$, $IQR = 1.00$; $U = 303.00$, $p = 0.0257$). We observed that software artifacts and hardware devices perceived as playable and isolated from the task often triggered episodes of attention loss. The design response to these observations is discussed in Section 7.1.

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